

NeuroPix: AI-Enhanced Image Processing and Management Web Application

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Abstract

NeuroPix is a browser application that brings conventional image controls, guided AI editing, comparison, and a personal gallery into one workflow. It connects OpenAI and locally hosted Stable Diffusion with cloud storage. Functional testing demonstrated the core editing and gallery workflows. The project aims to make these tasks simple and easy to access, although usability has not been measured against other tools.

Keywords: image processing, generative image editing, Flask, cloud storage, Stable Diffusion, web application

List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
AWS	Amazon Web Services
CI/CD	Continuous Integration and Continuous Delivery
EC2	Elastic Compute Cloud
GPU	Graphics Processing Unit
HLD	High-Level Design
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
RDS	Relational Database Service
S3	Simple Storage Service
SQL	Structured Query Language
SRS	Software Requirements Specification
SSH	Secure Shell
TLS	Transport Layer Security
UI	User Interface

Chapter 1 Introduction

1.1 Problem Statement and Significance

Combining conventional adjustments with AI-assisted changes can involve moving an image between tools. For example, a user may crop an image in one editor, upload it to an AI service, compare the result with the source, and organize the saved files manually. This creates extra upload, export, and file-selection steps. NeuroPix addresses this workflow by keeping the source, editing controls, comparison, and saved results together.

For users with basic editing needs, choosing tools and tracking file versions can take attention away from the image itself. An open-ended AI prompt also requires the user to describe both the desired change and what should remain unchanged. We wanted to reduce these steps through a focused browser interface and guided editing fields.

We developed NeuroPix for users with basic browser skills, without requiring professional editing experience. Users sign in to upload an image, apply Standard or AI edits, compare and download results, and manage their gallery in the browser. Figure 1.1 compares this workflow with using separate tools.

From separate tools to one editing workflow

The NeuroPix path keeps the source and results connected

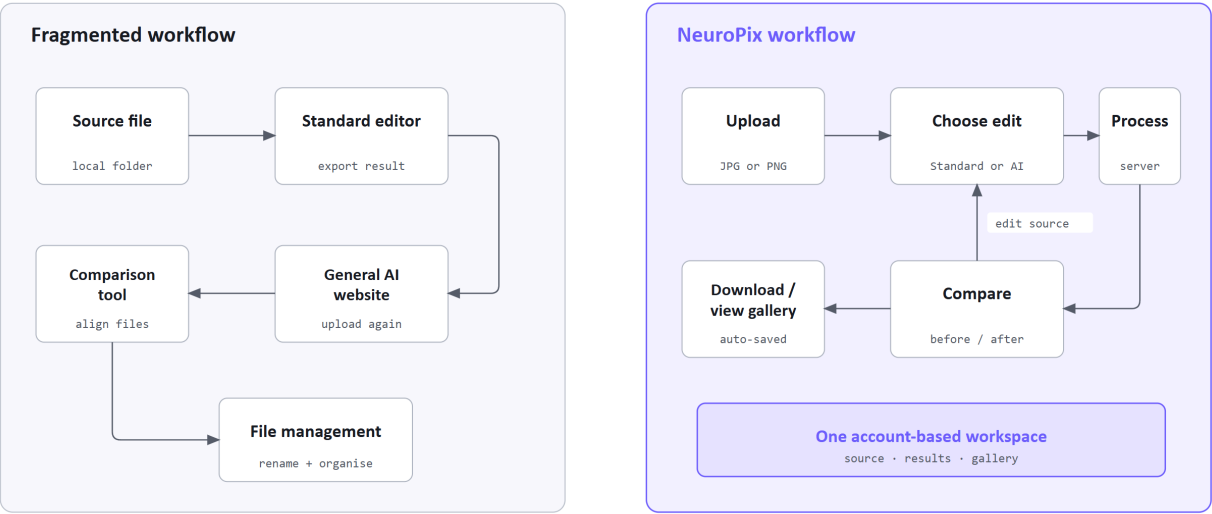


Figure 1.1: Fragmented and unified editing workflows.

The workflow requires file validation, account-specific gallery access, and coordination between image storage and database records. It must also handle AI services that may be slow or unavailable.

The project demonstrates how deterministic image operations and generative editing can share a simple account-based workflow. It also gives the team practical experience connecting web development, image processing, AI services, and cloud deployment.

1.2 Proposed Solution

NeuroPix is an authenticated web application that allows a user to upload an image, select either Standard or AI editing, inspect the processed result beside the original, download the output, and manage saved results in a personal gallery. Standard edits are processed with Pillow. AI edits use OpenAI by default, with locally hosted Stable Diffusion available as the self-hosted provider. NeuroPix provides separate fields for generative modification, background manipulation, and enhancement. It adds context to these instructions before sending an OpenAI request, so the user does not have to write one combined prompt. Image files are stored in Amazon S3, while MySQL stores the related account and image metadata [1]-[3].

1.2.1 Aim and Objectives

The main aim is to design, implement, deploy, and evaluate a web application that combines conventional image processing, AI-assisted editing, visual comparison, and user-specific image management.

The project objectives are:

- Provide an authenticated browser workflow for uploading and editing supported JPG and PNG images without requiring a desktop editor.
- Provide Standard controls and purpose-specific AI-editing fields, using OpenAI by default and self-hosted Stable Diffusion as the local alternative.
- Store original and processed image objects in Amazon S3 while storing account and image metadata in MySQL.
- Support before-and-after comparison, downloading, dashboard summaries, personal gallery management, and re-editing of original or edited images.
- Protect account and user-owned records through password hashing, parameterized queries, session checks, and ownership filtering.
- Evaluate the principal workflows through a compact automated test summary and six live use-case tests.

1.2.2 Project Deliverables

Table 1.1 lists the project deliverables and the evidence for each.

Table 1.1: Project deliverables and their evidence.

Deliverable	Evidence
Deployed web application	Authenticated dashboard, Studio, comparison, download, and gallery workflows available through the deployed site
Standard and AI editing	Pillow processing, OpenAI image editing, and the self-hosted Stable Diffusion integration
Persistence layer	Amazon S3 original/processed objects and MySQL user/image metadata
Design and deployment material	Current architecture, database, storage, sequence, and deployment diagrams based on the implementation
Verification material	Automated API/security summary plus six live use-case tests with sequence diagrams and screenshots
Final report	Project design, implementation, testing, and conclusions

1.3 Project Domain and Intended Users

NeuroPix combines conventional raster processing and generative image editing in an authenticated web application with cloud file storage. Users work in the browser, while the server handles processing and storage.

The intended users want straightforward adjustments or clear AI transformations without installing specialist software. They may be students, casual creators, or people trying AI editing. They need only common browser skills: creating an account, selecting files, adjusting controls, entering short descriptions, and choosing saved results.

Users access both AI providers through the same browser interface. They do not need to install the local model on their own devices, because it runs on the team's computer.

1.4 Scope and Limitations

The implemented scope includes account registration, session-based login, JPG/PNG upload validation, ten Standard editing controls, OpenAI prompt-based editing, self-hosted Stable Diffusion image-to-image processing, visual comparison, result download, dashboard summaries, and a user-filtered gallery. Gallery items can be reloaded for editing and deleted. Original and processed image files are kept in separate S3 locations, while the database stores ownership and editing metadata [1].

The project does not attempt to reproduce a professional layer-based editor. Video, vector graphics, RAW photography formats, animated images, real-time collaboration, social sharing, public image feeds, mobile-native applications, and foundation-model training are outside the scope. It also does not claim enterprise-scale availability or performance because those properties have not been measured.

Several limitations follow from the chosen design. Generative results are probabilistic and may change details that the user did not mention. OpenAI processing depends on valid credentials, internet access, service availability, latency, and account limits. The local provider depends on the RTX 3080 computer, its Docker service, and the reverse connection being available. Processing is synchronous, so a slow AI request keeps the user waiting for the current request to finish. Operations that affect both S3 and MySQL also do not have complete transactional rollback, so a partial external failure can require manual correction.

1.5 Definitions

Term	Meaning in NeuroPix
Standard edit	A deterministic image operation performed with Pillow, such as crop, rotation, brightness, contrast, blur, sharpness, or grayscale
AI edit	A prompt-based image transformation processed by OpenAI or the self-hosted Stable Diffusion service
Structured pre-prompting	Server-side addition of field-specific rules, preservation guidance, and source-image context around the user's OpenAI request
Image-to-image generation	Generative processing that uses an uploaded image as the visual starting point for a new result
Object storage	Storage of image files as objects in Amazon S3 rather than as binary data in MySQL
Metadata	Database information about a user-owned image, including object locations, edit type, and related attributes
Session	Signed session data identifying the logged-in user and the image staged for editing
Comparison slider	Interface control that reveals portions of the original and processed image in the same viewing area

Chapter 2 Background and Related Work

2.1 Image-Editing Background

NeuroPix combines deterministic and generative editing. Its Standard mode uses Pillow for operations such as crop, rotation, brightness, contrast, saturation, detail adjustment, and partial grayscale [4]. These controls use measurable settings and follow a fixed processing order. Generative editing interprets a text instruction and may alter content beyond the requested region. Latent diffusion makes this type of image generation practical and is the basis of the Stable Diffusion family used by the self-hosted provider [5].

Hosted and self-hosted AI differ in where generation runs and who maintains it. A managed service such as OpenAI provides image editing through an API [6]. A self-hosted model runs on hardware maintained by the application team, giving the team control over the service but responsibility for its resources and availability.

2.2 Related Systems

Adobe Photoshop is a professional image editor with detailed selection, layer, compositing, correction, and generative tools. Its generative features can add, remove, or modify content within a mature editing environment [7]. These tools support a wider range of professional tasks than the focused editing workflow we set out to build.

Canva provides a browser-based design environment with photo editing integrated into a broader content-creation workflow [8]. Its templates and design tools extend beyond the image-editing tasks in this project. We chose a smaller scope for NeuroPix so we could concentrate on combining Standard and AI editing with comparison and a personal gallery.

2.3 Comparative Analysis

Table 2.1: Comparison of NeuroPix with selected related systems.

Category	Adobe Photoshop	Canva	NeuroPix
Main problem and users	Detailed professional image creation and manipulation	Accessible visual content creation for a broad user base	Focused Standard and AI image editing for authenticated browser users
Access and design approach	Desktop-centered professional workspace with extensive tools	Browser-based template and design workspace	Browser-based upload, edit, compare, download, and gallery path
Editing and output	Deep raster editing, layers, selections, and generative features	Photo edits within designs and templates	Ten Standard controls, OpenAI editing, self-hosted Stable Diffusion, and direct image output
Guidance and management	Professional controls and workflow features	Guided design interface and asset organization	Purpose-specific prompt fields, automatic OpenAI context, comparison, and personal gallery
Relevant limitation	More complexity than this project's intended task flow	Broader design scope than the project's focused image workflow	Narrower toolset, synchronous processing, and environment-dependent AI services

2.4 NeuroPix Differentiation

NeuroPix is designed for simplicity and easy access. A user signs in, uploads one image, chooses a mode, reviews the result, and downloads it or manages it through the gallery. The interface keeps the source and processed result connected in a focused workspace. We have not measured its usability against other tools. Compared with entering a single prompt in a general-purpose AI website, NeuroPix offers separate modification, background, and enhancement fields. It adds scope and preservation guidance automatically [1], so users can describe the desired change without composing all of that context themselves. The application uses the selected provider's model without claiming to improve its capabilities.

Chapter 3 System Analysis

3.1 Requirements Identification Method

We checked the final implementation against our SRS and HLD. The routes, service modules, database schema, and interface showed which requirements were implemented, while the deployment files identified the operating requirements. Where the planning documents differed from the finished system, this chapter describes the implemented behavior [1]-[3].

Table 3.1: Sources used to recover and confirm the final requirements.

Evidence source	Purpose and use
Flask routes and service modules	Identified authentication, upload, processing, download, gallery, and provider behavior implemented by the backend
Frontend files and live interface	Confirmed page access rules, visible controls, browser validation, feedback, comparison, and navigation
Database schema and persistence code	Confirmed entities, ownership links, stored fields, queries, and deletion behavior
Deployment files and live environment	Confirmed EC2 hosting, Nginx, HTTPS, RDS, S3, automated restart, health checking, and the reverse-SSH path
SRS and HLD	Recorded project intent and earlier design. The implementation took precedence where they differed

Table 3.2 identifies the main inputs and expected outputs for each editing mode.

Table 3.2: Principal system inputs, processing, and outputs.

Mode	Inputs	Processing	Expected output
Standard editing	Supported image and numeric settings for crop, rotation, brightness, contrast, exposure, saturation, blur, sharpness, and grayscale	Validation followed by the fixed Pillow pipeline	Processed JPEG, comparison view, downloadable result, and saved metadata
OpenAI editing	Supported image plus any combination of generative, background, and enhancement instructions	Structured prompt construction, output-size selection, hosted image request, and decoded result	AI-edited PNG, comparison view, download, and gallery record
Local AI editing	Supported image, provider selection, and nonempty user text	Combined prompt sent to the self-hosted Stable Diffusion image-to-image service	Locally generated image when the service and reverse connection are available

3.2 Stakeholders and User Characteristics

NeuroPix users need to select and edit images with basic browser skills, sliders, and short written instructions. The interface should show when a request is processing or has failed. Users also expect their gallery records to remain separate from other accounts.

The project team designs, implements, deploys, tests, and documents the system. The supervisor and evaluators review whether the application and evidence satisfy the project goals. Processing, storage, and service availability depend on OpenAI, AWS, and the computer running the local model.

3.3 Functional Requirements

Table 3.3 groups the functional requirements by account access, editing, persistence, and image management.

Table 3.3: Grouped functional requirements.

Requirement group	Implemented behavior
Account access	Register unique usernames with passwords of at least eight characters and store salted hashes. Support login, authenticated sessions, current-user lookup, and logout. Redirect signed-out users and protect private API routes
Upload and Standard editing	Accept JPG, JPEG, or PNG within orientation-aware dimension limits and stage one image. Apply the ten Standard controls in order. Display, compare, download, and save the result
OpenAI editing	Accept three purpose-specific text fields and an upscale choice. Remove empty fields and add scope, preservation, and source-image context. Select a supported output size, submit the request, and decode the image
Local Stable Diffusion	Report local-provider availability and allow provider selection. Combine nonempty user text and call the image-to-image service. Return the generated image or report connection or generation failure
Persistence	Store original and processed objects in Amazon S3. Store ownership, object locations, edit type, and upload time in MySQL. Retrieve owned records and remove the selected record and objects
Dashboard and gallery	Show edit counts, recent activity, and only the current user's images. Support comparison, reloading an owned original or edited image, and deletion after confirmation

3.4 Nonfunctional Requirements

Table 3.4: Nonfunctional requirements and current design response.

Quality area	Requirement and design response
Security	Hash passwords, use parameterized SQL, require sessions for private operations, filter records by user ownership, keep secrets in server configuration, and serve the deployed site through HTTPS
Usability	Use consistent navigation, clear labels, immediate status messages, purpose-specific AI fields, and a visual comparison instead of requiring the user to inspect separate files
Reliability and compatibility	Validate files and settings, report external-provider failures, keep a health endpoint, and support the current desktop Chrome workflow. AI availability depends on external services
Maintainability and performance	Separate routing, editing, storage, database, and AI code. Use environment-based configuration and automated deployment checks. Synchronous AI processing can keep requests open for an extended period

3.5 Use-Case Model and Descriptions

Figure 3.1 places the six principal use cases inside the NeuroPix boundary and their supporting services outside it. OpenAI and local Stable Diffusion provide AI processing, while S3 and MySQL support persistence. Only selected service links are drawn because editing and gallery cases share them. Registration and login do not require an existing session.

NeuroPix use-case model

Editing and gallery actions require login; registration and login do not

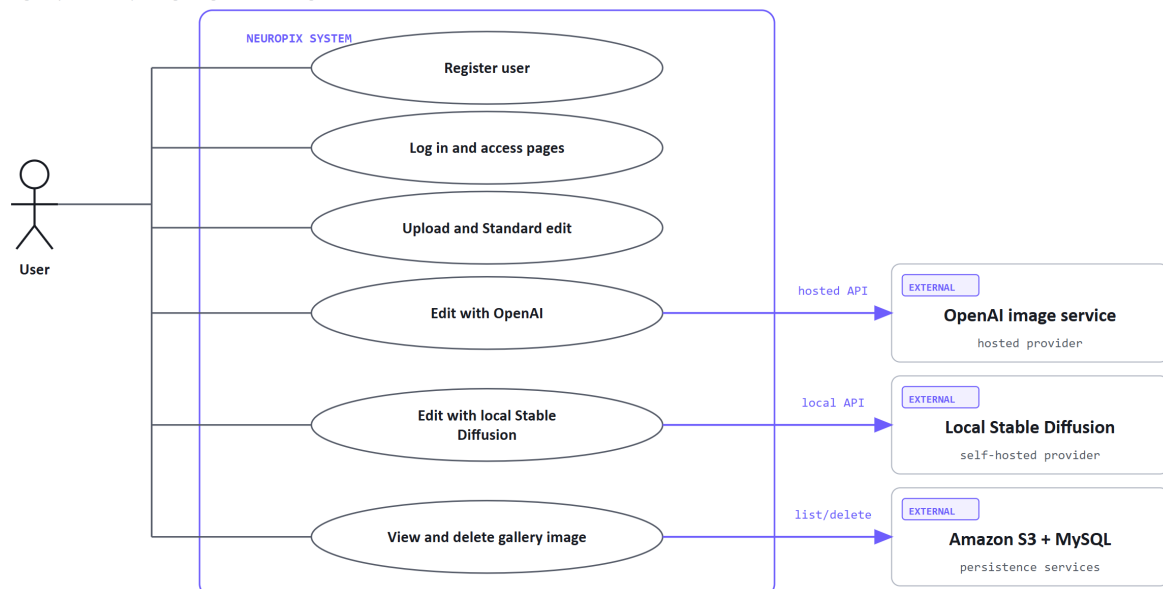


Figure 3.1: NeuroPix use-case model.

Table 3.5: Compact descriptions of the six principal use cases.

Use case	Preconditions and trigger	Main flow, alternatives, and postcondition
UC-T01 Register	User is logged out and opens registration	User enters a unique username and matching valid password. The server checks uniqueness, hashes the password, and creates the account. Invalid data or an existing name is rejected. A successful user can proceed to login.
UC-T02 Login	Account exists and user opens login	User submits valid credentials. The server verifies the stored hash and creates a session. Invalid credentials are rejected. Success opens the protected dashboard.
UC-T03 Standard edit	User is authenticated and has a supported image	User uploads the image, selects Standard settings, processes it, and compares the result. Invalid inputs stop the request. Success stores the original, output, and metadata and enables download.
UC-T04 OpenAI edit	User is authenticated, image is staged, and OpenAI is configured	User selects OpenAI and enters one or more purpose-specific instructions. NeuroPix constructs the request and stores the returned result. Provider errors are displayed, with no new output.
UC-T05 Local AI edit	User is authenticated, image is staged, and the local service is available	User selects Local Stable Diffusion and submits text. The deployed backend forwards the request through the configured local path. Connection or generation failure is reported. A successful request stores the returned result.
UC-T06 Gallery delete	User is authenticated and owns at least one saved image	Gallery lists the owned record. After confirmation, the server verifies ownership, removes the database record and related S3 objects, and refreshes the gallery. A missing or foreign record is rejected.

3.6 Current System Workflow

After authentication and upload, users choose Standard or AI editing. Both branches share storage, comparison, and download. From comparison, users can return to mode selection and edit the same staged source without uploading again. To use a processed result as the next source, they reload its edited version from the gallery. Figure 3.2 shows this successful workflow. Validation and provider failures leave the user on the current page with a status message.

End-to-end editing activity

One uploaded source can be reused for Standard and AI edits

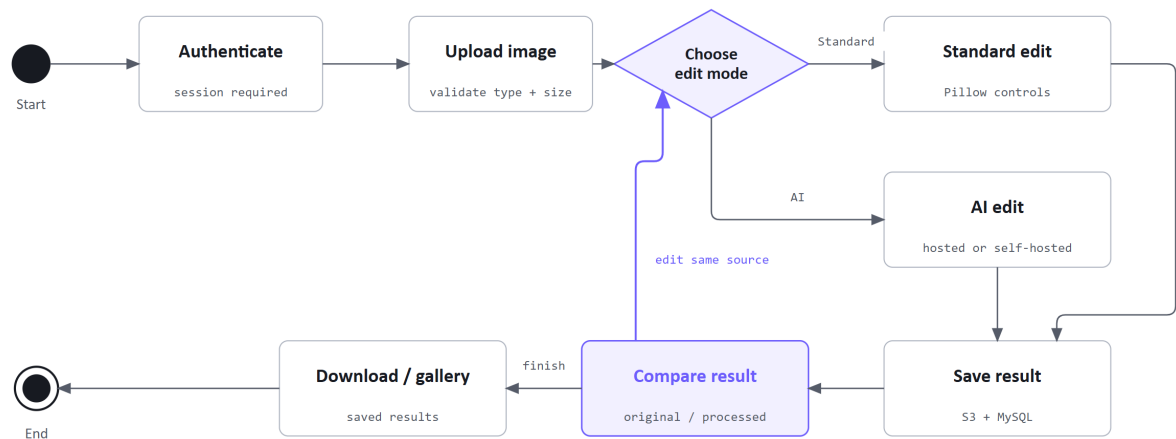


Figure 3.2: Authenticated editing activity.

3.7 Data Requirements and Business Rules

The system has two persistent entities. The Users table stores authentication details, and the Images table stores ownership and the locations of uploaded and processed objects. Section 4.7 shows their relationship in the ER diagram.

Table 3.6: Users entity data dictionary.

Field	Type and rule
UserID	Integer primary key generated by the database
Username	Required unique username used at login
PasswordHash	Required salted password hash. The original password is not stored
CreatedAt	Account creation timestamp

Table 3.7: Images entity data dictionary.

Field	Type and rule
ImageID	Integer primary key generated by the database
UserID	Required foreign key to Users. Deleting a user also deletes their image records
OriginalFilePath	Required S3 object location for the uploaded source
ModifiedFilePath	Optional S3 object location for the processed result
EditType	Enumerated value identifying Standard or AI editing
UploadDate	Record creation timestamp used by dashboard and gallery ordering

The main business rules are that usernames are unique, private operations require an authenticated session, every image record belongs to exactly one user, and gallery queries include that ownership condition. The schema permits an empty processed location, while a completed edit records both object paths. Deletion is restricted to an owned record.

3.8 Software and Hardware Requirements

The implementation has three main environment requirements:

- Users need a current desktop browser, internet access, and an image within the supported format and dimension limits.
- The production application requires Python, Flask, Pillow, MySQL connectivity, S3 access, provider configuration, a Linux EC2 instance, Nginx, and a valid TLS certificate.
- The self-hosted provider requires a Windows computer with Docker, the downloaded Stable Diffusion resources, enough storage and memory, and the project's NVIDIA RTX 3080.

The EC2 application can operate independently of the local computer. Local generation requires that computer and its connection to the deployed application to remain active.

Chapter 4 System Design

4.1 Design Approach

NeuroPix uses a layered web design. The frontend is responsible for navigation, input collection, local preview, and visible status. Flask handles authentication and coordinates requests. Separate modules perform deterministic editing, AI-provider calls, database access, and object storage. This division was selected to keep provider credentials and persistence logic on the server while allowing the browser interface to remain simple.

The design also separates persistent files from relational records. S3 stores image objects, while MySQL stores small searchable records that connect each image to its owner. The backend can use either OpenAI or the self-hosted service without changing the browser workflow. OpenAI is the default.

4.2 High-Level Architecture

Figure 4.1 separates the user's browser, the EC2 application host, and external services. Nginx receives HTTPS requests and forwards them to Flask. Flask calls the editing modules and the image service, which coordinates RDS metadata and S3 files. Provider calls and storage credentials remain on the server.

NeuroPix high-level architecture

Browser interaction, application services, and persistence are separated

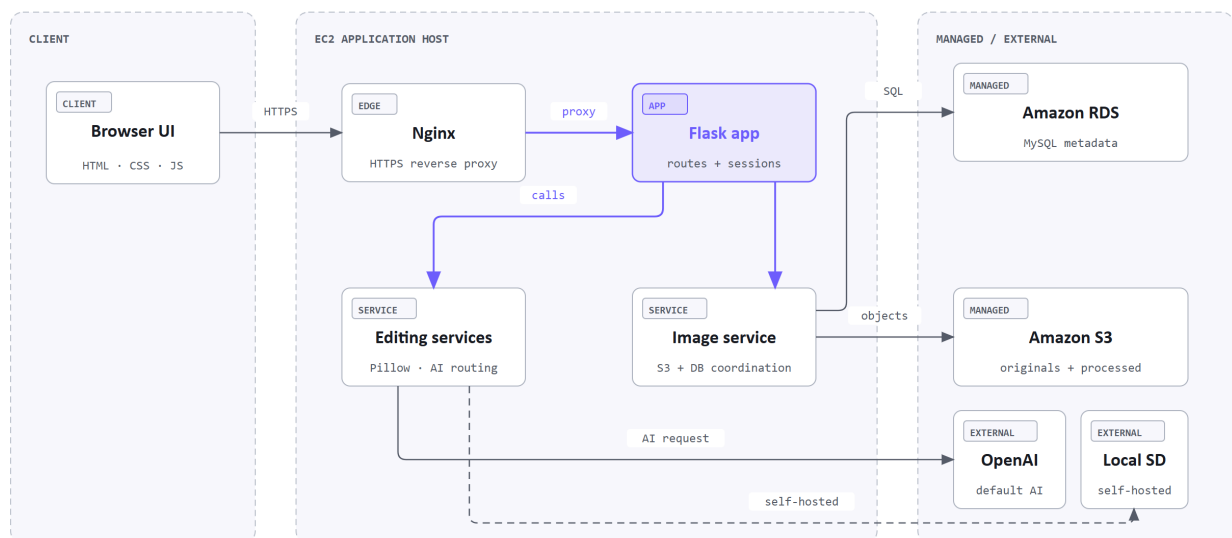


Figure 4.1: NeuroPix system architecture.

The shared backend lets the interface present the same comparison and gallery features for results from Pillow, OpenAI, or the local model. Processing remains synchronous, so the application holds each request open until processing and storage finish or an error is returned.

4.3 Frontend Navigation and Access Boundaries

The public pages introduce the project and provide registration or login. Successful authentication opens the dashboard and makes the Studio and Gallery available. The browser redirects signed-out users to login. Private API requests are protected by backend session and ownership checks.

Frontend navigation and access boundaries

Public entry pages and shared signed-in navigation

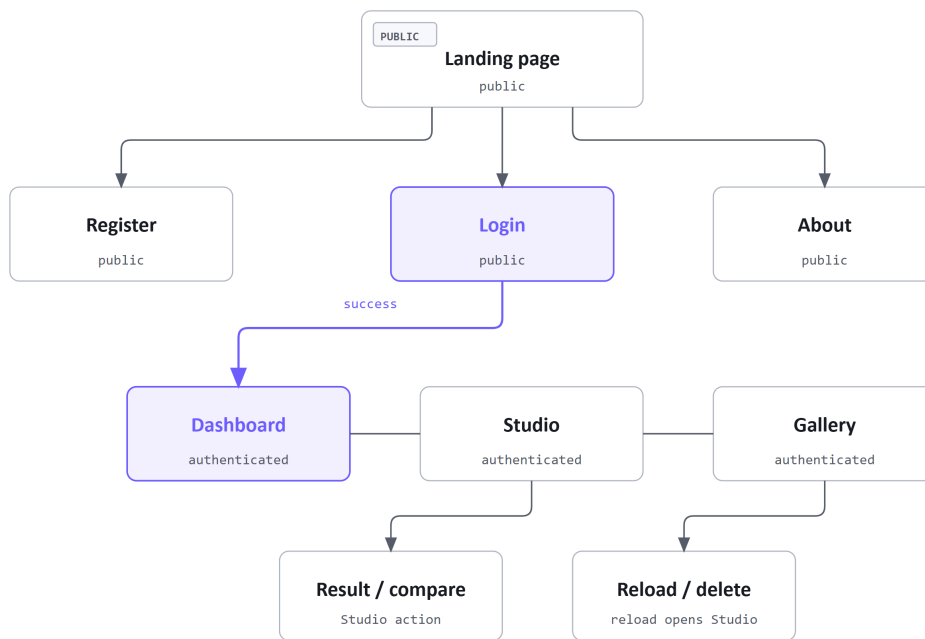


Figure 4.2: Frontend navigation and access.

The Studio keeps upload, mode selection, processing status, result viewing, and download in one page. The Gallery is separate because it operates on already saved records. The sidebar links the Dashboard, Studio, and Gallery directly. Reloading an original or edited gallery image opens it in the Studio.

4.4 Backend and API Design

The backend exposes a small set of page and JSON routes. Authentication and gallery routes are user-specific. Upload prepares the current source, process selects the Standard or AI service, and download returns the most recent processed file in the active session.

Table 4.1: Implemented backend route groups.

Route or group	Method	Responsibility
/ and /health	GET	Serve the public entry point and report application health
/api/auth/register	POST	Validate registration, hash the password, and create the user
/api/auth/login, /api/auth/logout, /api/auth/me	POST/GET	Verify credentials, manage the session, and return the active identity
/api/upload	POST	Validate a supported image and stage it temporarily
/api/process	POST	Select Standard, OpenAI, or local processing and persist the result
/api/download	GET	Return the latest processed file associated with the current session
/api/gallery	GET	List the current user's stored image records
/api/gallery/<id>/load	POST	Verify ownership and stage the selected original or edited image
/api/gallery/<id>	DELETE	Verify ownership and remove the selected record and related objects
/api/local-model/status	GET	Report whether the self-hosted endpoint is reachable

The backend also validates input checked by the frontend. JSON errors use clear messages and suitable status codes so the browser can display the failure without exposing internal exception details.

4.5 Standard Image-Processing Design

The Standard editor follows the fixed order in Figure 4.3. Each operation acts on the previous result, so changing the order can change the output. Cropping before rotation keeps crop percentages tied to the source orientation. Brightness and exposure share one factor because both adjust image brightness.

Implemented Standard processing order

Each operation receives the output of the preceding operation



Figure 4.3: Standard processing order.

Percentages from the interface are converted into factors expected by Pillow. Crop width and height describe retained proportions rather than coordinates. The grayscale control blends the color and grayscale images, which supports partial desaturation as well as a fully grayscale result. The route saves the completed image as JPEG.

4.6 AI Prompting and Provider Routing

The OpenAI path is designed around three user intentions: generative modification, background manipulation, and enhancement. Empty fields are removed. Each remaining field receives a scope rule, and the request also receives shared preservation guidance and the source aspect ratio. NeuroPix calculates a provider-supported output size before sending the image and constructed prompt. This is the project's structured pre-prompting approach.

Structured prompting and AI-provider routing

OpenAI adds scope and preservation guidance. Local combines user text.

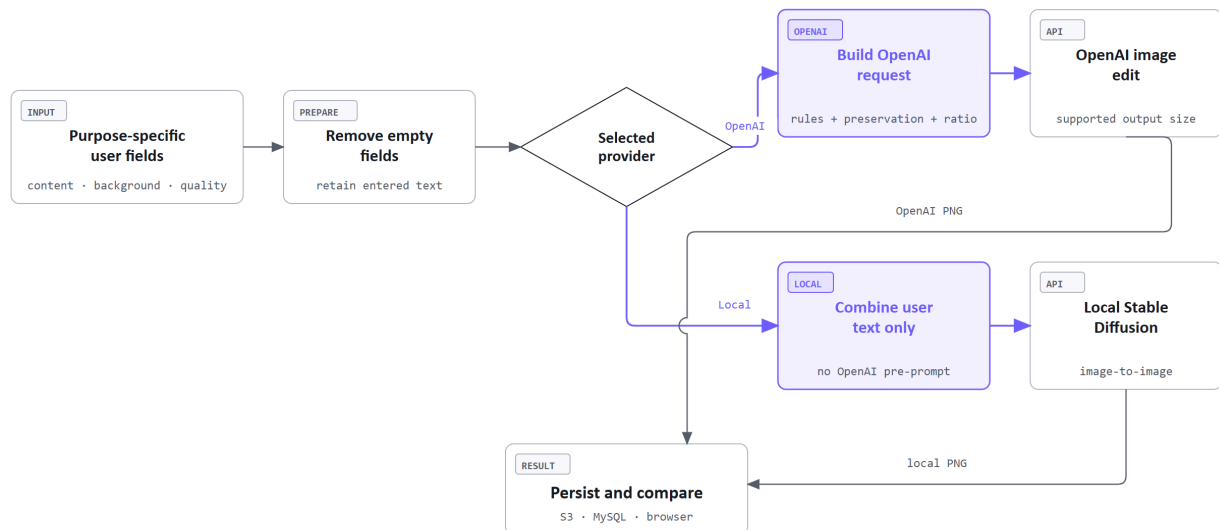


Figure 4.4: AI prompt and provider routing.

Users can write short instructions because the backend adds guidance for each field. For example, it marks text in the background field as a background change and asks the model to preserve unmentioned content. The model may still alter those details, so preservation is not guaranteed.

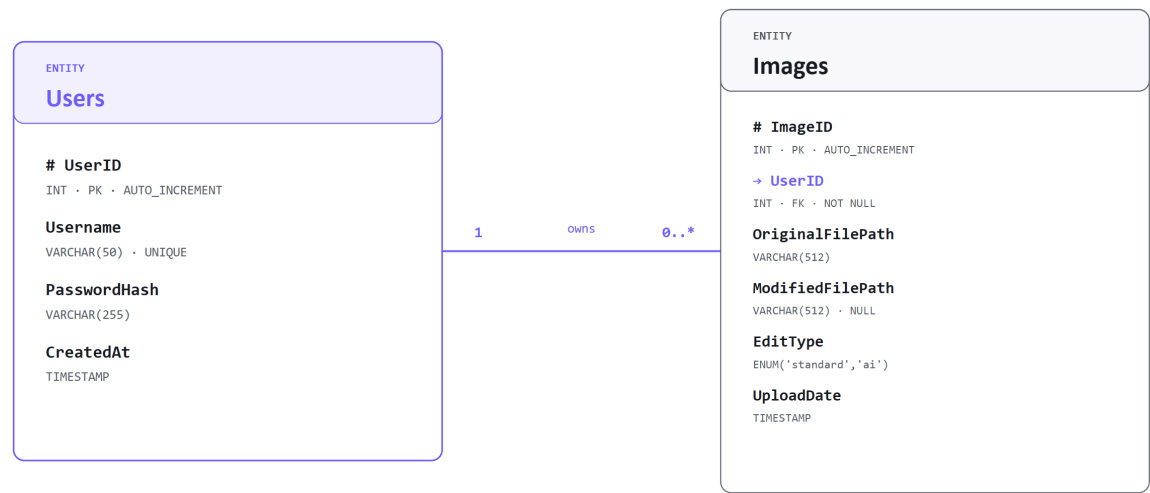
The local path combines the user's nonempty text and forwards the staged image to the self-hosted endpoint. **Only the OpenAI** path adds scope and preservation rules. Both provider paths return a processed image to the same comparison and storage workflow.

4.7 Database Design

A user can have no saved images or many, but each image record must have exactly one owner. Figure 4.5 shows this relationship, enforced by UserID. The field definitions are listed in Section 3.7.

NeuroPix entity-relationship model

Image files remain in S3; the database stores ownership and object paths



Foreign-key cascade removes image records, not S3 objects.

Figure 4.5: NeuroPix data model.

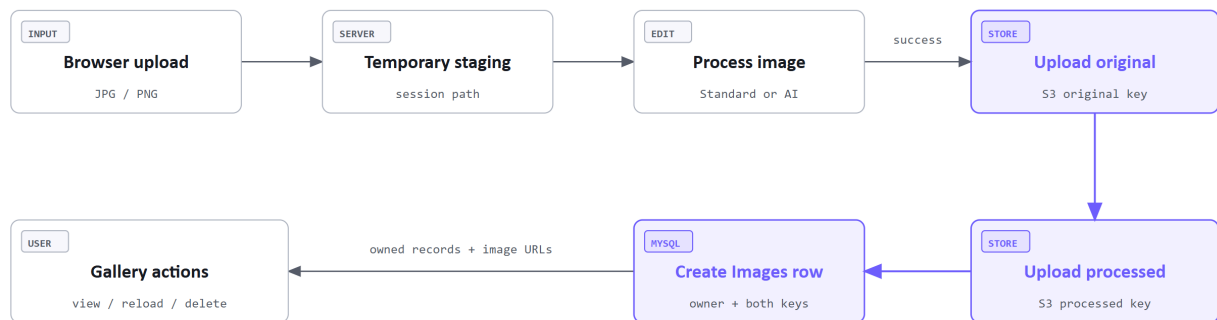
The username unique constraint supports account identity, and the foreign key prevents an image record from referring to an unknown user. The database, hosted in Amazon RDS for MySQL [9], stores paths rather than binary image data. This keeps relational queries focused on identity, ownership, edit type, and ordering.

4.8 Image Storage and Lifecycle

An upload prepares a source for editing without creating a gallery record. Figure 4.6 shows how a successful edit becomes a saved source/result pair. The record keeps both versions available for comparison and later editing, while ownership links the pair to one account.

Image storage and lifecycle flow

Both files are saved after processing succeeds



Reload stages the chosen original or edited file. Gallery display uses image URLs.

Delete removes the owned database row, then targets its S3 objects. Partial failures need recovery.

Figure 4.6: Image storage lifecycle.

S3 and MySQL do not participate in one shared transaction. The application orders operations and reports failures, but a failure between services can still leave an object or record that needs manual correction. Recovery is therefore a limitation of the current storage design.

4.9 Security Design

Table 4.2: Security controls and remaining boundaries.

Area	Implemented control	Remaining boundary
Credentials	Werkzeug salted password hashing, with no original passwords stored	Password reset and multi-factor authentication are not implemented
Database access	Parameterized queries and uniqueness/foreign-key constraints	Service credentials still require careful environment management
Session and ownership	Login-required routes and UserID filters for gallery load/delete	Session security depends on production secret and cookie configuration
Files and providers	Server-side validation and server-held API/storage credentials	AI providers receive the image required for the selected service
Network	HTTPS termination through Nginx and Certbot-managed certificate	Local-model availability also depends on the reverse tunnel

The design applies authorization at the route and query level instead of trusting a gallery identifier supplied by the browser. This matters because hiding another user's record in the interface would not be sufficient protection by itself.

4.10 Design-Level Interaction Sequences

The authentication sequence follows a new account through registration and login. Creating an account does not sign the user in. The backend creates a session only after a successful password check.

Authentication design sequence

New account registration followed by a separate login

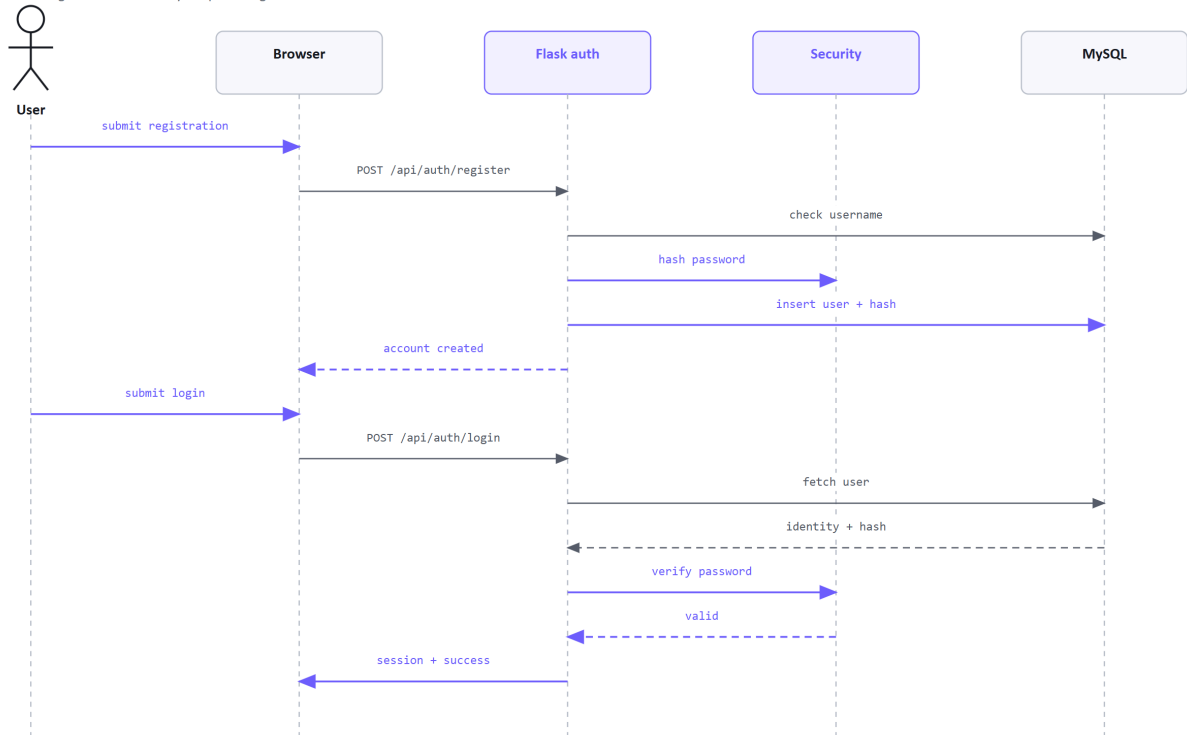


Figure 4.7: Authentication sequence.

The Standard sequence separates upload from processing. Pillow returns an image, which Flask saves as JPEG before persistence. Further edits reuse the staged source.

Upload and Standard-processing design sequence

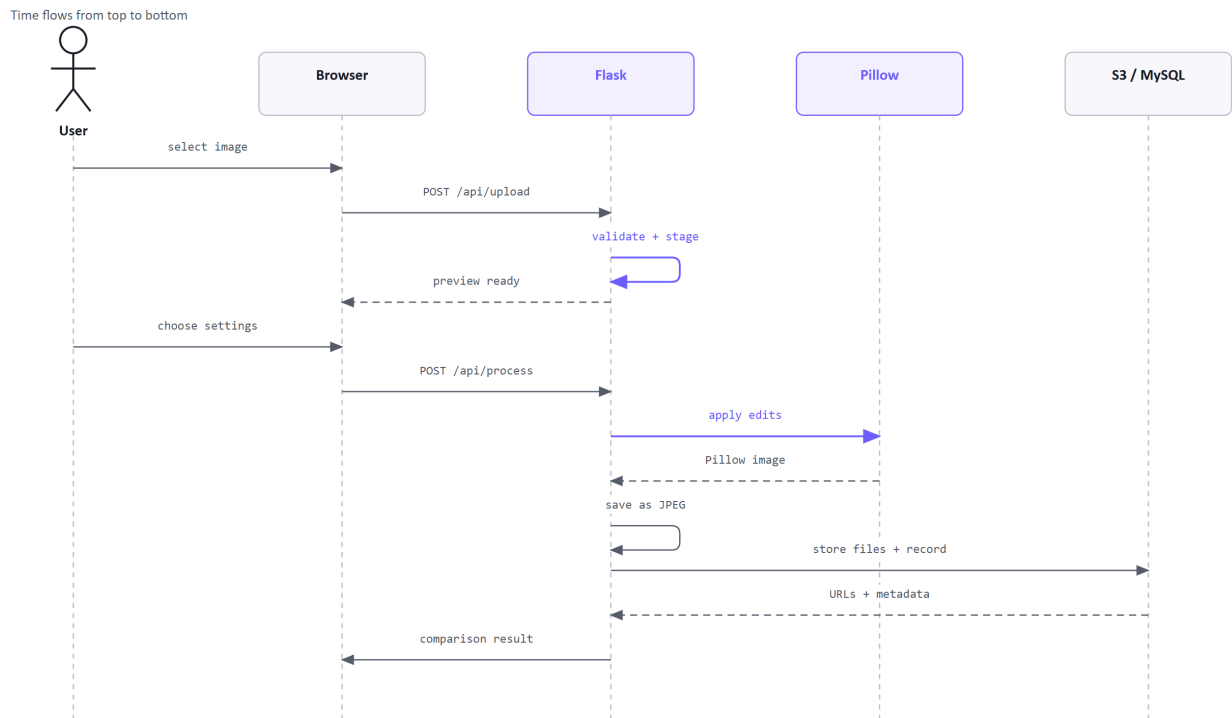


Figure 4.8: Standard-edit sequence.

The Flask backend selects one AI provider per request. Its AI module builds the OpenAI prompt or combines local-model instructions, then the route stores the result and returns image URLs.

AI-provider alternatives design sequence

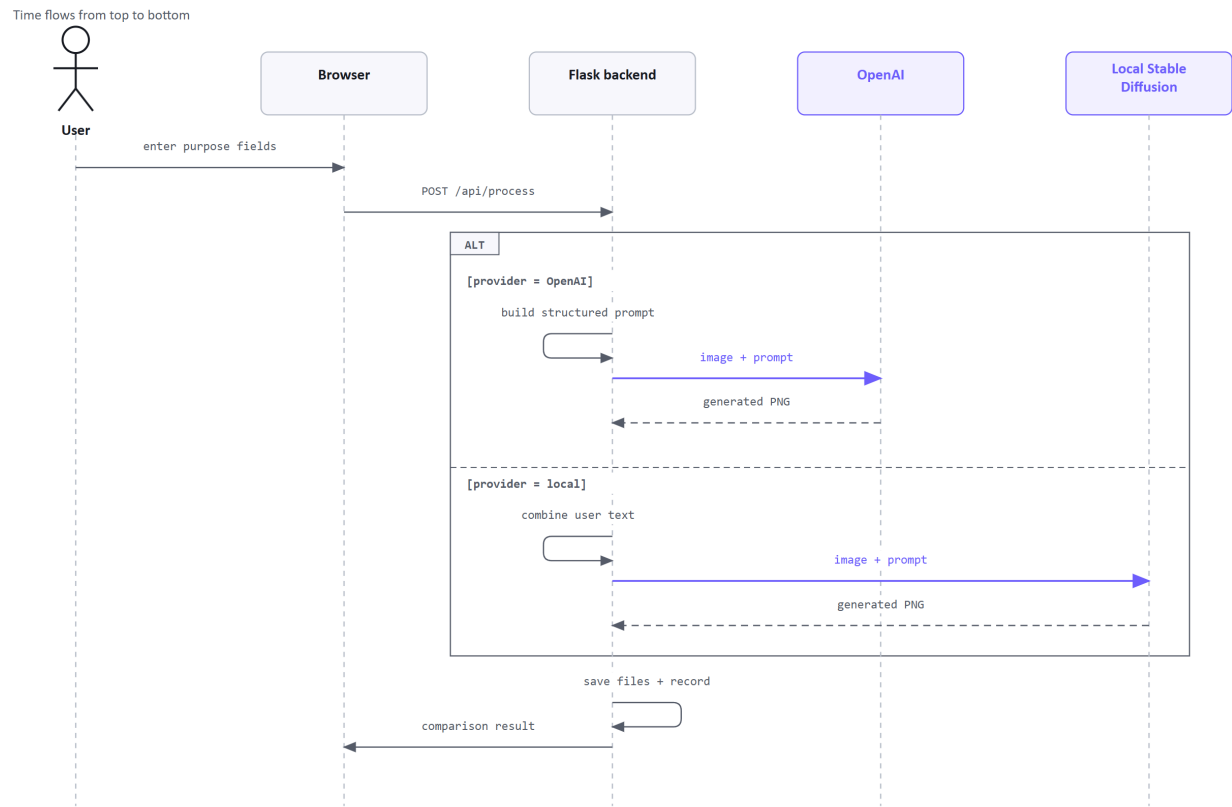


Figure 4.9: AI-provider sequence.

Gallery queries return only owned records. After checking ownership, reload stages the selected image version. Deletion removes the database row and then targets its S3 objects.

Gallery operations design sequence

Time flows from top to bottom

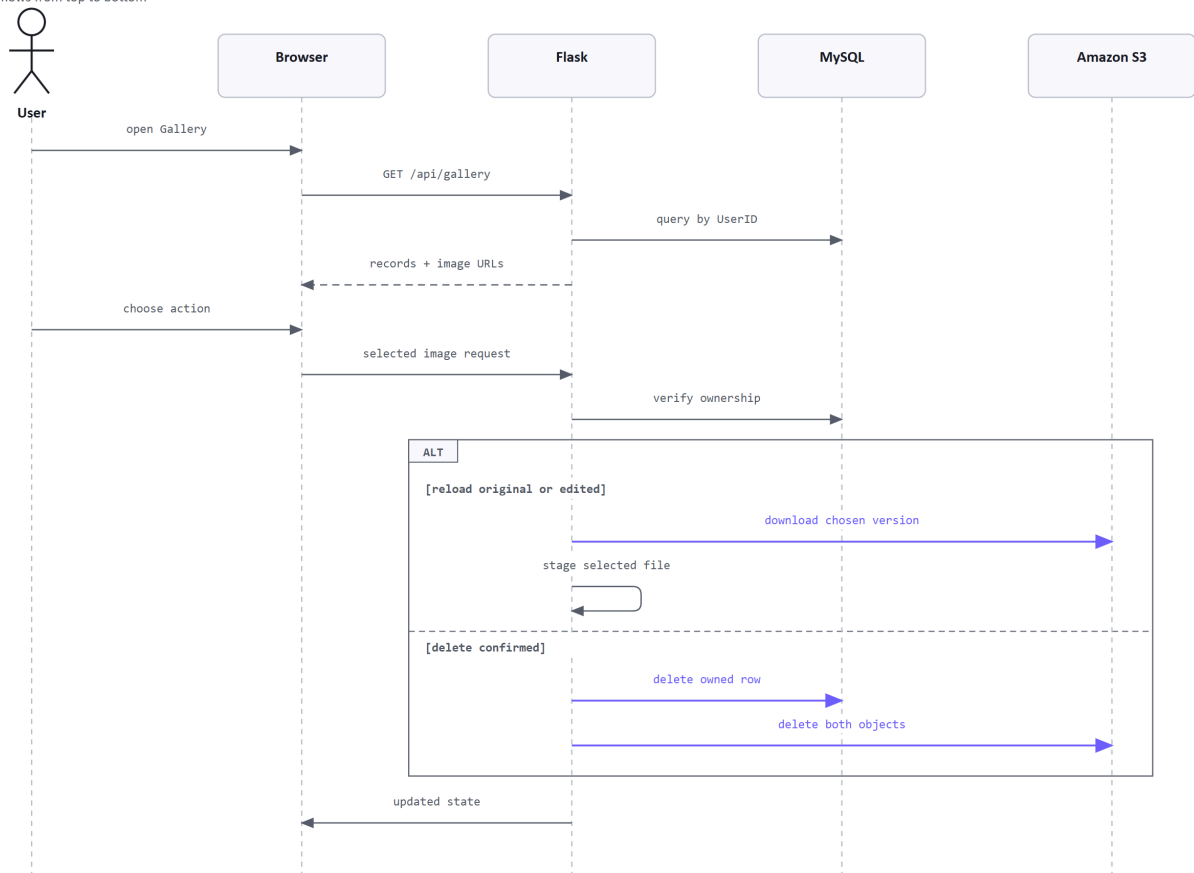


Figure 4.10: Gallery sequence.

4.11 Deployment and Reverse-SSH Design

The public domain resolves to the EC2 host. Nginx terminates HTTPS using the Certbot-managed certificate and forwards application requests to the Flask service. Flask connects to Amazon RDS for MySQL and Amazon S3 using server-side configuration. A self-hosted GitHub runner on the EC2 instance updates the application after changes on main, restarts the service, and checks /health.

Production deployment and local-model path

Windows opens the tunnel to EC2. Editing requests travel back to the local model.

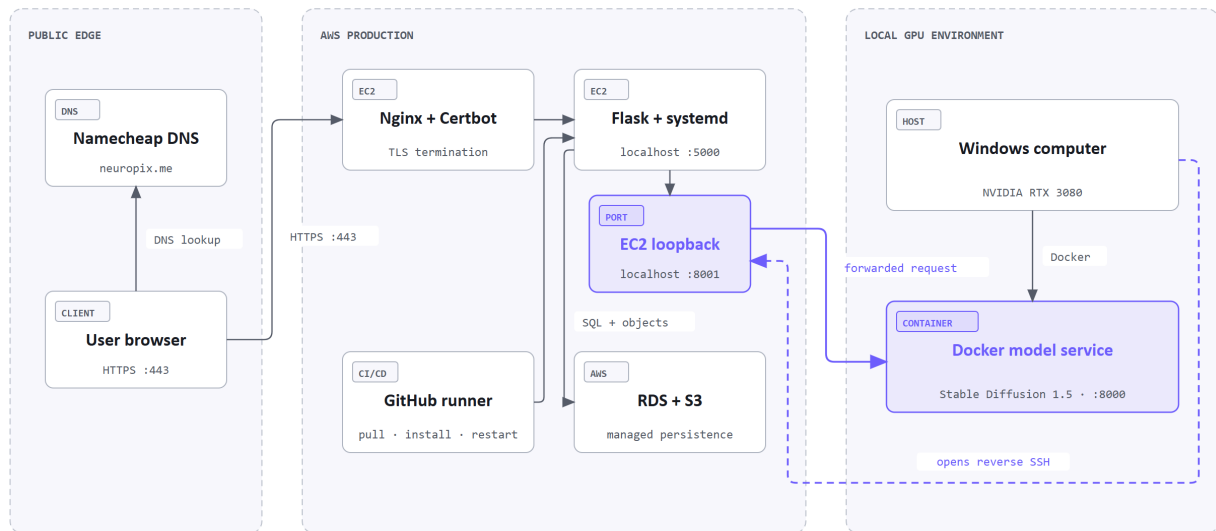


Figure 4.11: Production deployment and reverse SSH.

The self-hosted model is outside the public cloud environment. Its Windows computer runs the Dockerized service on the RTX 3080 and creates a reverse SSH tunnel to the EC2 host. Flask calls the EC2-side tunnel endpoint, which forwards the request to the local service. This avoids exposing the home computer directly, but local generation is unavailable whenever the computer, Docker service, internet connection, or tunnel is offline. OpenAI therefore remains the default provider.

4.12 Development Decisions and Trade-Offs

Cost was our main constraint. We also considered what we could learn and how much support was available. Table 4.3 summarizes the choices and their trade-offs.

Table 4.3: Principal development decisions and trade-offs.

Decision	Alternatives considered	Reason and trade-off
AWS deployment	Smaller or less common hosting services	The student trial reduced initial cost. AWS had extensive documentation, and experience with a widely used provider was useful for our CVs. We still had to stay within the trial limits.
Namecheap domain	Domain registration through AWS	The AWS trial did not cover domain registration. We found Namecheap's free student offer while looking for ways to reduce costs outside AWS.
OpenAI as default AI provider	Local-only generation	OpenAI had just released a new image model. Its managed service, ease of use, and extensive documentation made it our default, with usage costs and dependence on an external service.
Local Stable Diffusion 1.5	Camber, Lightning.ai, and cloud GPU hosting	Cloud GPU trials used credit while the GPU was on and were awkward for repeated testing and tuning. Stable Diffusion fit our RTX 3080 and let us learn to deploy a model on our own hardware. Local generation depends on that computer being available.
Direct work on main	Feature branches and pull requests	Two teammates were new to Git and GitHub. One branch simplified our small project and reduced workflow mistakes, but provided less isolation and formal review.

Keeping costs down required us to look across student programs, cloud trials, and equipment we already owned. We combined AWS hosting and storage with a free student domain, OpenAI as the default provider, and a model running on our own computer.

Chapter 5 Implementation

5.1 Technology Stack and Environment

Table 5.1 summarizes the technologies used to implement the browser interface, processing services, storage, and deployment.

Table 5.1: Implemented technology stack.

Layer	Technology and role
Frontend	HTML, CSS, and browser JavaScript for pages, controls, validation, requests, previews, comparison, and status messages
Backend	Python and Flask for routing, sessions, validation, service coordination, and JSON responses
Standard processing	Pillow for crop, rotation, tonal/color controls, detail controls, grayscale blending, and image encoding
Hosted AI	OpenAI image service for the default generative editing path
Local AI	Stable Diffusion 1.5 image-to-image served through LitServe in Docker on an NVIDIA RTX 3080
Metadata	MySQL in Amazon RDS, accessed through parameterized backend queries
Image files	Amazon S3 originals and processed objects
Production hosting	Amazon EC2, Nginx, systemd, Certbot, Namecheap DNS, and a GitHub self-hosted deployment runner
Verification	Pytest for API/security checks and Chrome automation at a 1920 × 1080 viewport for live use cases

5.2 Frontend and Authentication Implementation

The public pages share a consistent design and link to separate registration and login forms. JavaScript submits the credentials to the backend and displays the returned result. Registration checks matching passwords in the browser for immediate feedback, while the server independently enforces required fields, minimum password length, and username uniqueness. The stored value is a Werkzeug-generated password hash.

Login verifies the submitted password against that hash and stores the user identifier and username in the Flask session. The Dashboard, Studio, and Gallery use a browser-side identity check to redirect signed-out users. The backend independently checks the session on private API requests and filters image records by ownership. The frontend requests the current user's details to display who is signed in. Logout clears the session and returns the user to the public site.

The private pages share a sidebar so the user can move between the overview, editing workspace, and gallery without returning through the public navigation. The dashboard first displays cached gallery data, then refreshes the counts and recent records from the user's gallery API.

5.3 Upload and Standard Processing Implementation

The Studio accepts file selection or drag and drop. The frontend checks the file type, size, and image dimensions before upload. The backend checks the filename extension, opens the image with Pillow, and reads its dimensions. It applies limits of 1920×1080 for landscape and 1080×1920 for portrait. Server-side validation is necessary because browser checks can be bypassed.

After a valid upload, the source is staged temporarily and available to the editing controls. Standard settings are submitted as JSON. The Pillow editing module applies the pipeline in Figure 4.3. The route then calls the image service to coordinate S3 uploads and the database record. It returns the image URLs to the workspace, and the saved edit appears on the dashboard and in the gallery.

The comparison modal uses the original and processed images in one frame with a draggable boundary. This is useful for grayscale, crop, and generative edits because the user can inspect the same area without opening two browser tabs. Download returns the most recently processed file using the session state established by the successful request.

5.4 AI Editing Implementation

Selecting AI mode changes the controls without changing the uploaded source. OpenAI is selected by default, using gpt-image-2.5-sunburst unless configured otherwise. The browser collects generative modification, background manipulation, enhancement, and upscale values. The backend ignores empty text fields, adds the structured context described in Section 4.6, and selects an output size compatible with the source aspect ratio. It sends the source image and final prompt to the configured OpenAI image model and decodes the returned image bytes as PNG.

When Local Stable Diffusion is selected, the frontend can check `/api/local-model/status` before processing. The backend combines the nonempty user text and sends the image and prompt to the configured local endpoint with a five-minute request timeout. The service runs Stable Diffusion 1.5 on the RTX 3080 with ten inference steps, guidance scale 7.5, and strength 1.0.

Both AI providers return to the same comparison, download, S3, and MySQL workflow. Provider-specific failures are converted into visible errors, and the browser does not treat an earlier image as a newly successful result.

5.5 Persistence, Dashboard, and Gallery Implementation

Database access is contained in dedicated functions that use placeholders rather than building SQL from user input. Account queries look up a unique username. Image queries include the current UserID, so the database filters records by ownership during retrieval, reload, and deletion. The interface is not responsible for this check.

The S3 service generates separate keys for original and processed objects. Metadata records retain both locations so the application can present a processed result while allowing either version to be reloaded for another edit. The dashboard aggregates Standard and AI counts and selects recent records. The gallery lists the same owned records in date order and marks AI-edited images visually.

Deletion first verifies that the record belongs to the current user. The implementation removes the database row and then targets the stored objects. Chapter 6 checks the visible deletion workflow. Section 4.8 describes the remaining recovery limitation.

5.6 Production Deployment and Continuous Delivery

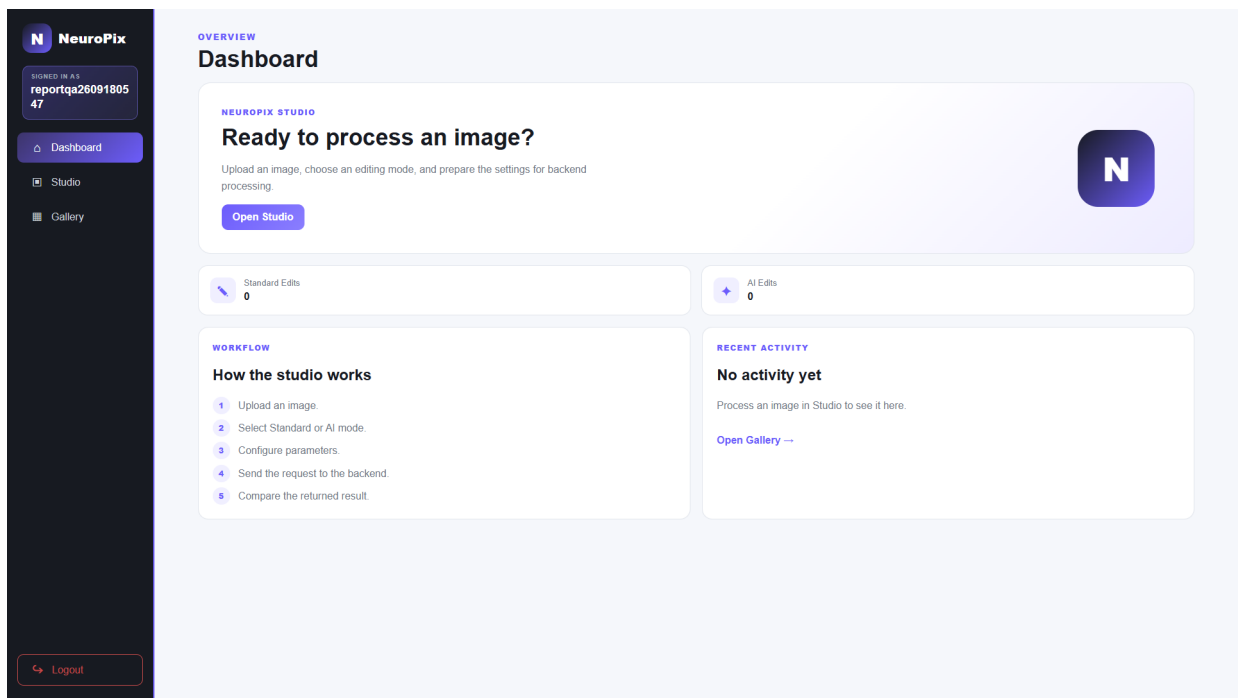
The deployed components have separate responsibilities:

- Namecheap DNS points the domain to EC2, where Nginx terminates HTTPS using the Certbot certificate and forwards requests to the Flask service.
- Environment values provide database, S3, session, and provider configuration without placing credentials in the repository or frontend.
- The GitHub workflow uses a self-hosted EC2 runner to pull the application state, install requirements, restart the systemd service, and call the health endpoint.
- The local-model service runs in Docker on the Windows computer. The computer opens the reverse SSH tunnel used by the production application to reach that service. Local editing requires the tunnel and service to stay active.

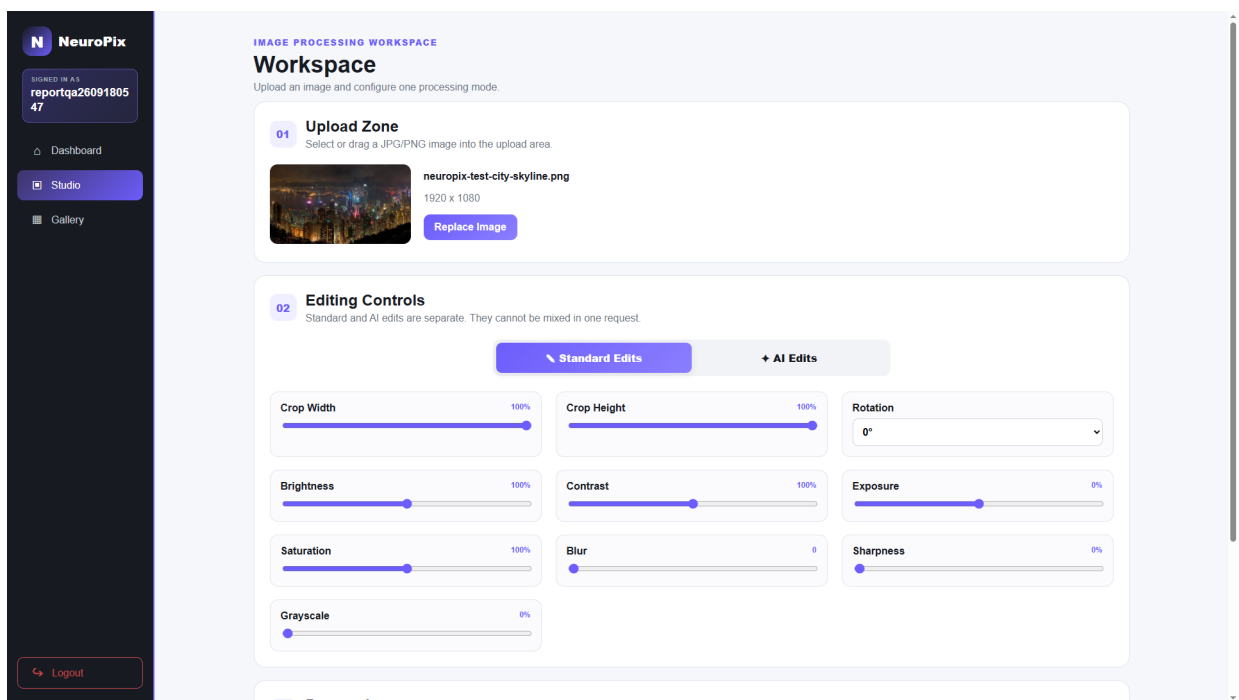
This deployment process can be repeated and reports a failure if the restarted application does not answer its health check. Figure 4.11 shows how the self-hosted provider joins the production system.

5.7 Implemented Input and Output Screens

Figure 5.1 shows the protected dashboard and the Studio after the controlled city-skyline image was uploaded. The sidebar and section numbering provide a consistent path from overview to editing.



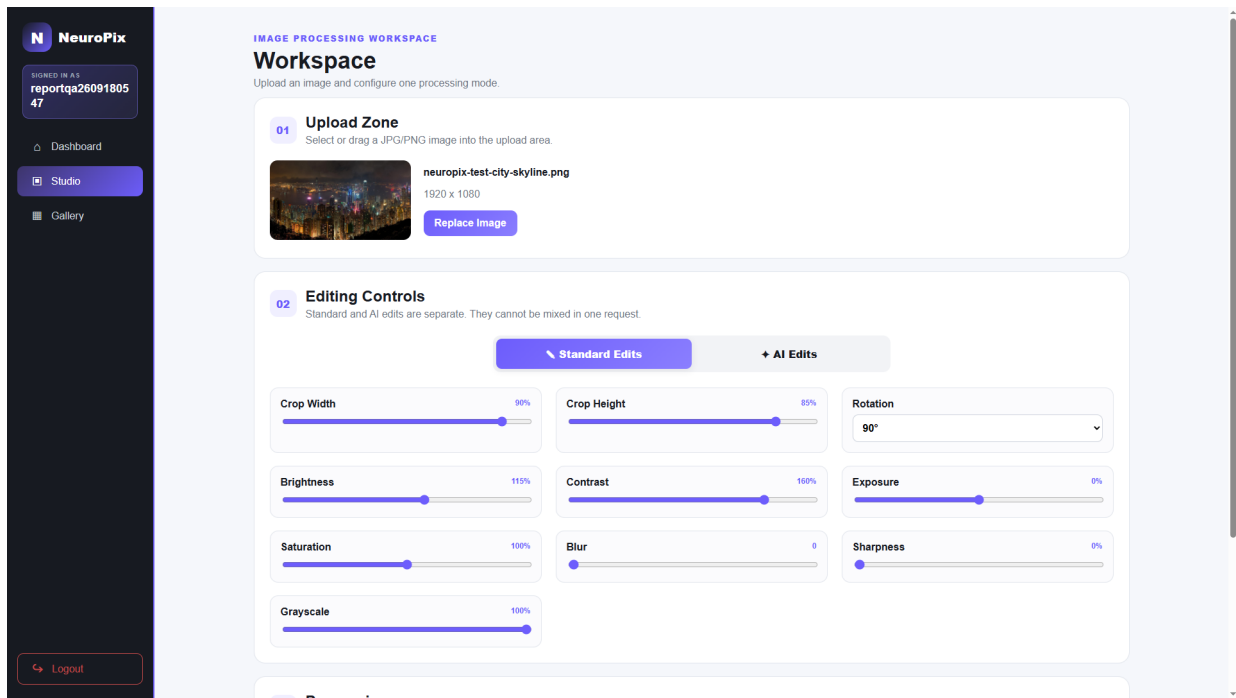
5.1 (a) Dashboard



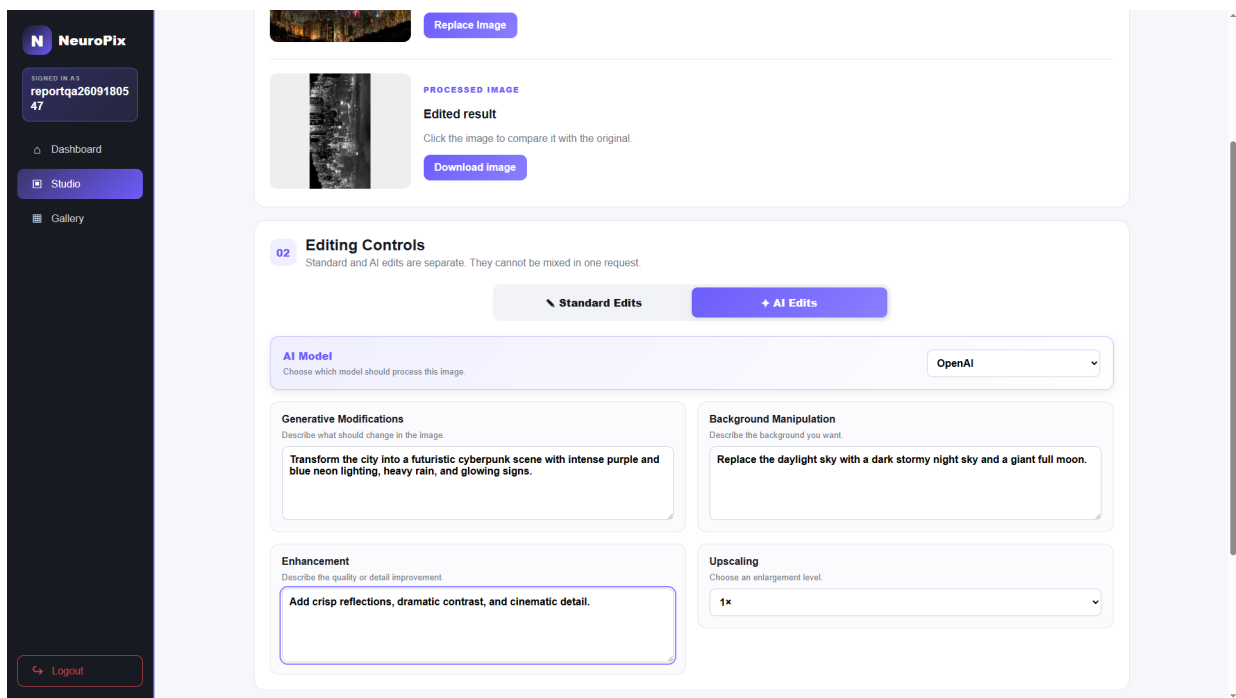
5.1 (b) Studio upload workspace

Figure 5.1: Dashboard and Studio upload.

The Standard controls expose numeric settings, while the AI panel replaces them with provider and purpose-specific text fields. The interface shows the controls for the selected mode, keeping Standard settings separate from AI instructions.



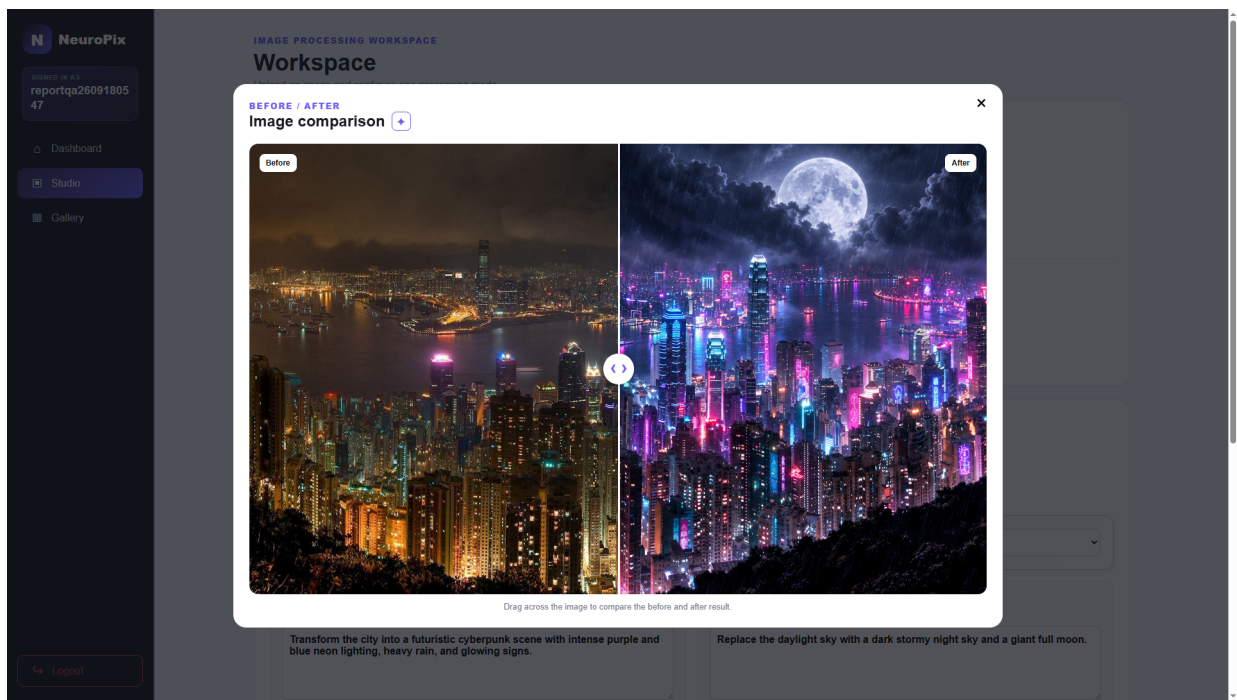
5.2 (a) Standard editing controls



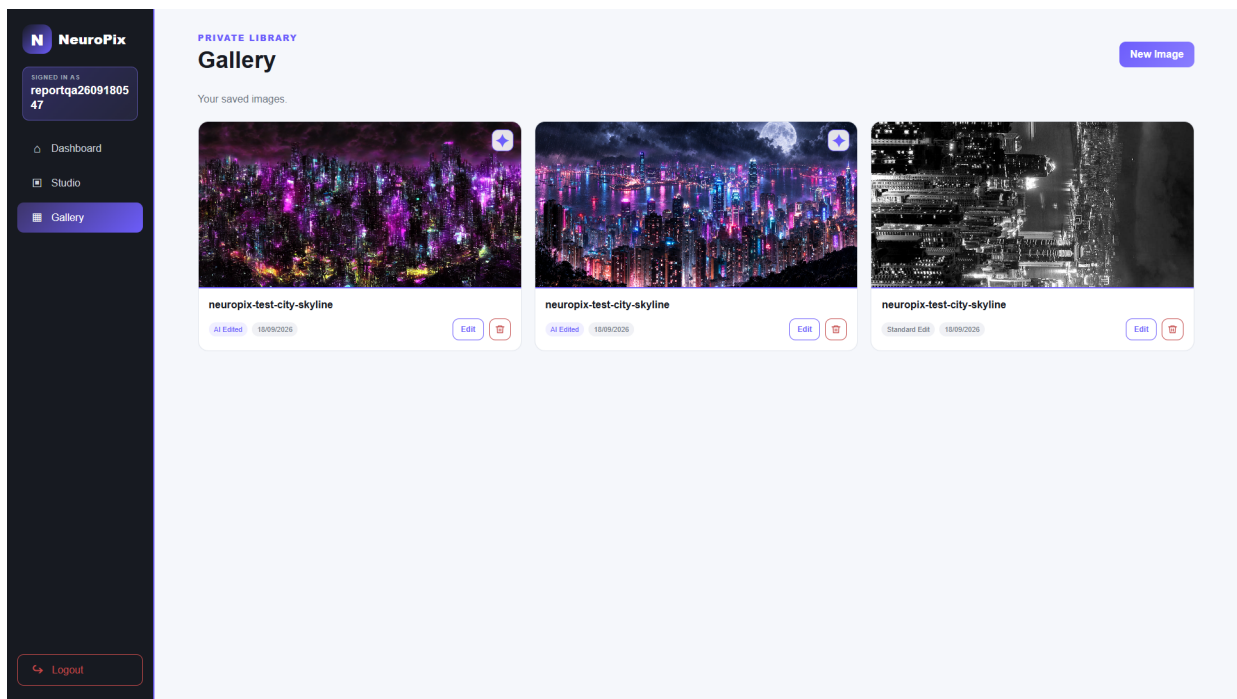
5.2 (b) OpenAI editing controls

Figure 5.2: Standard and OpenAI controls.

The processed output can be compared directly with its source, and saved results appear in the private gallery with their edit type and management actions.



5.3 (a) Before-and-after comparison



5.3 (b) Personal gallery

Figure 5.3: AI comparison and gallery.

Chapter 6 Testing and Evaluation

6.1 Testing Strategy

Testing focused on the main user workflows. A compact automated suite checked selected API and security behavior, while six live use cases exercised the deployed site in Chrome. Each use case defined preconditions and expected results, then recorded the observed actions and outcome. Sequence diagrams explain the interactions, and screenshots provide evidence of the visible behavior.

The automated suite was run from the project repository on September 18, 2026. It covered authentication, upload validation, processing responses, password hashing, selected local-provider behavior, and mocked storage coordination. These tests do not establish cross-account gallery isolation or verify the actual SQL queries.

Table 6.1: Automated API and security test result.

Test scope	Command and result
API and security checks	<code>python -m pytest -q tests/test_api.py tests/test_security.py.</code> All 28 tests passed in 3.97 seconds

6.2 Live Test Environment and Controlled Data

Table 6.2: Live use-case test environment.

Item	Verified configuration
Site	<code>https://neuropix.me</code> production deployment over HTTPS
Browser	Google Chrome controlled through Playwright CLI
Viewport	1920 × 1080 at normal browser zoom
Execution date	September 18, 2026
Account	New dedicated test account created through UC-T01
Image	<code>neuropix-test-city-skyline.png</code> , PNG, 1920 × 1080
Standard settings	Crop 90% × 85%, rotation 90°, brightness 115%, contrast 160%, and grayscale 100%
AI instruction	Extreme cyberpunk night transformation with neon lighting, rain, a stormy sky, a large moon, reflections, and cinematic detail

The city skyline was chosen because geometry, grayscale, lighting, background, and generative changes are easy to recognize. The same staged image was reused after UC-T03 so UC-T04 and UC-T05 did not repeat login and upload.

6.3 Evidence and Completion Rules

A case passed when we observed its expected visible results. Screenshots record the interface, and sequence diagrams explain the implementation. This evidence alone cannot confirm every internal storage or database operation. All cases required an image to be returned. For gallery deletion, the selected card had to remain absent after refresh.

Testing was complete when all six cases had a recorded outcome supported by live evidence. Each result covers only the stated preconditions and actions. Other features and failure paths need separate tests.

6.4 UC-T01: Register a New User

Precondition: the test username does not exist and the browser has no authenticated session. The user opens registration, enters a unique username and matching valid password, and submits the form. The expected result is successful account creation followed by the login page.

UC-T01 registration sequence

Time flows from top to bottom

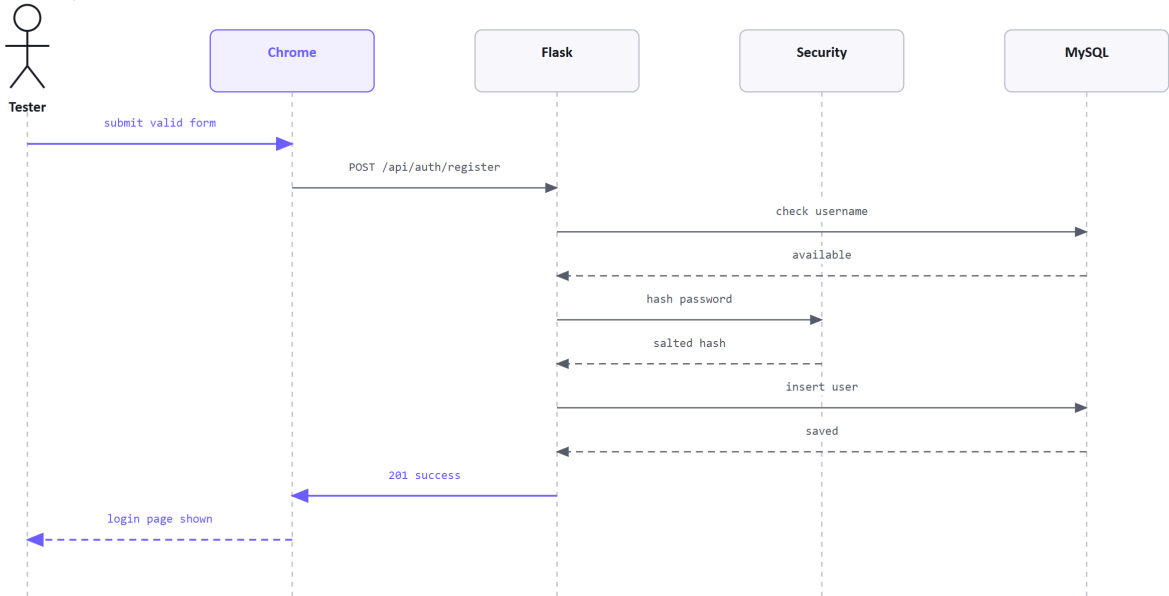


Figure 6.1: UC-T01 registration sequence.

Table 6.3: UC-T01 execution result.

Step	Observed result
Open registration	Registration form displayed with username, password, and confirmation fields
Enter valid data	Username remained visible and both password fields remained masked
Submit	The site reported that the account was created and redirected to login
Final status	Pass

CREATE ACCOUNT

Join NeuroPix

Create your account to access the workspace.

Username

Choose a username

Password

Minimum 8 characters

Confirm Password

Repeat your password

Create Account

Already have an account? [Login](#)

UC-T01 (a) Registration form

CREATE ACCOUNT

Join NeuroPix

Create your account to access the workspace.

Username

reportqa2609180547

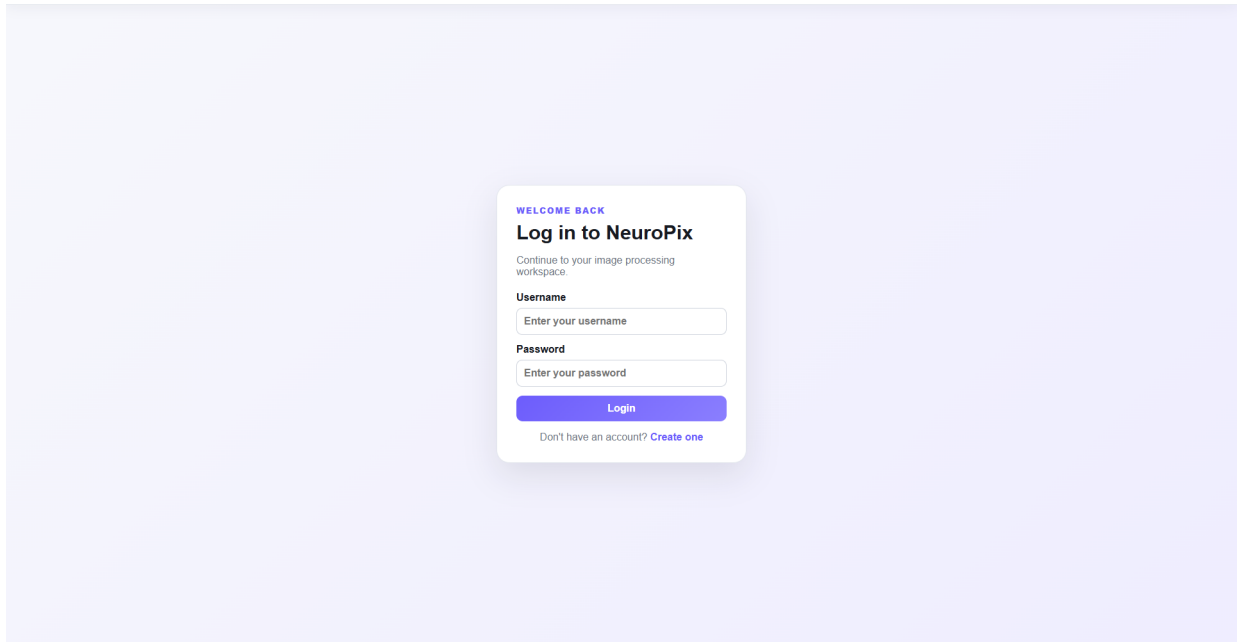
Password

Confirm Password

Create Account

Already have an account? [Login](#)

UC-T01 (b) Registration details entered



UC-T01 (c) Account created and login page opened

Figure 6.2: UC-T01 registration evidence.

6.5 UC-T02: Login and Access a Protected Page

Precondition: the UC-T01 account exists and the user is on the login page. The user enters valid credentials and submits. The expected result is an authenticated session and access to the dashboard.

UC-T02 login and dashboard access sequence

Time flows from top to bottom

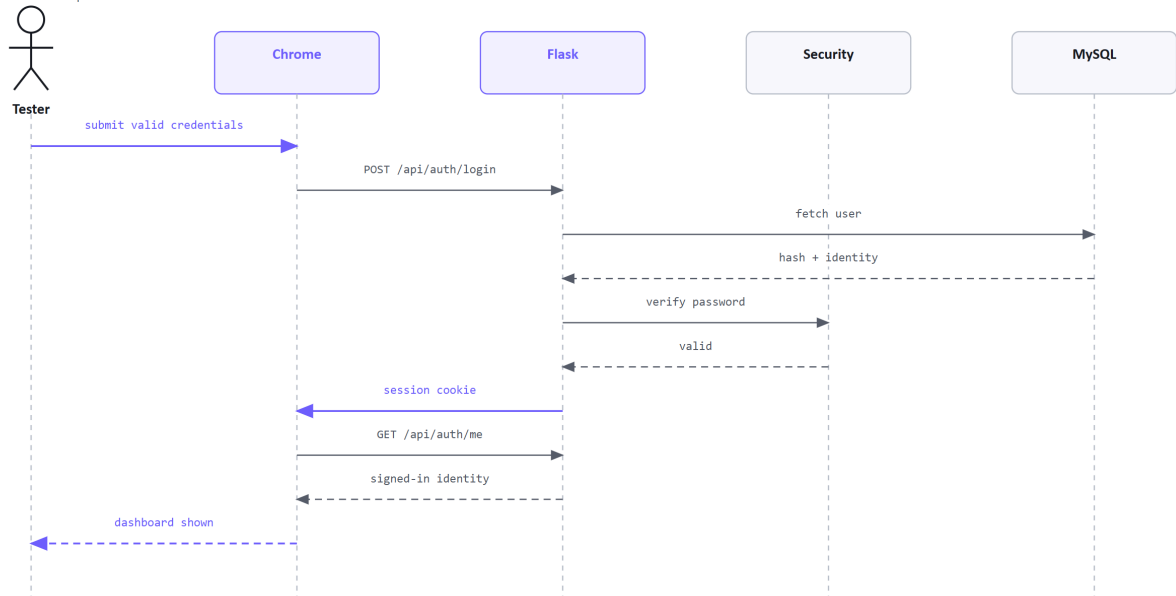
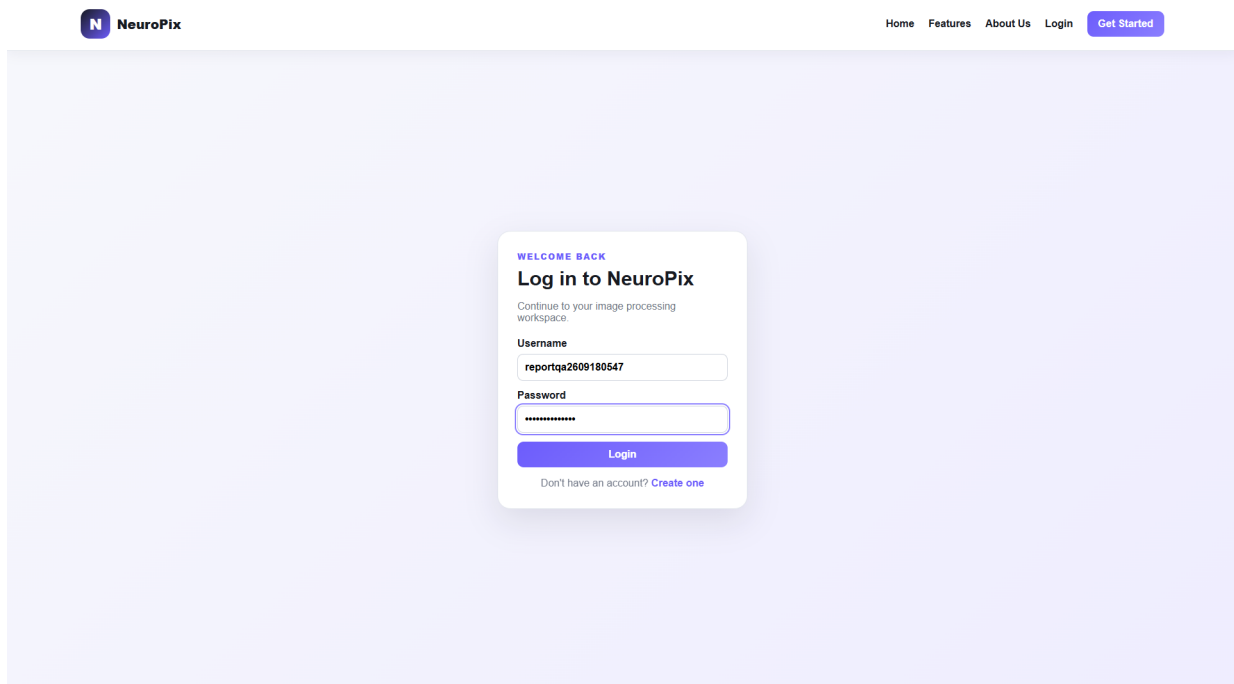


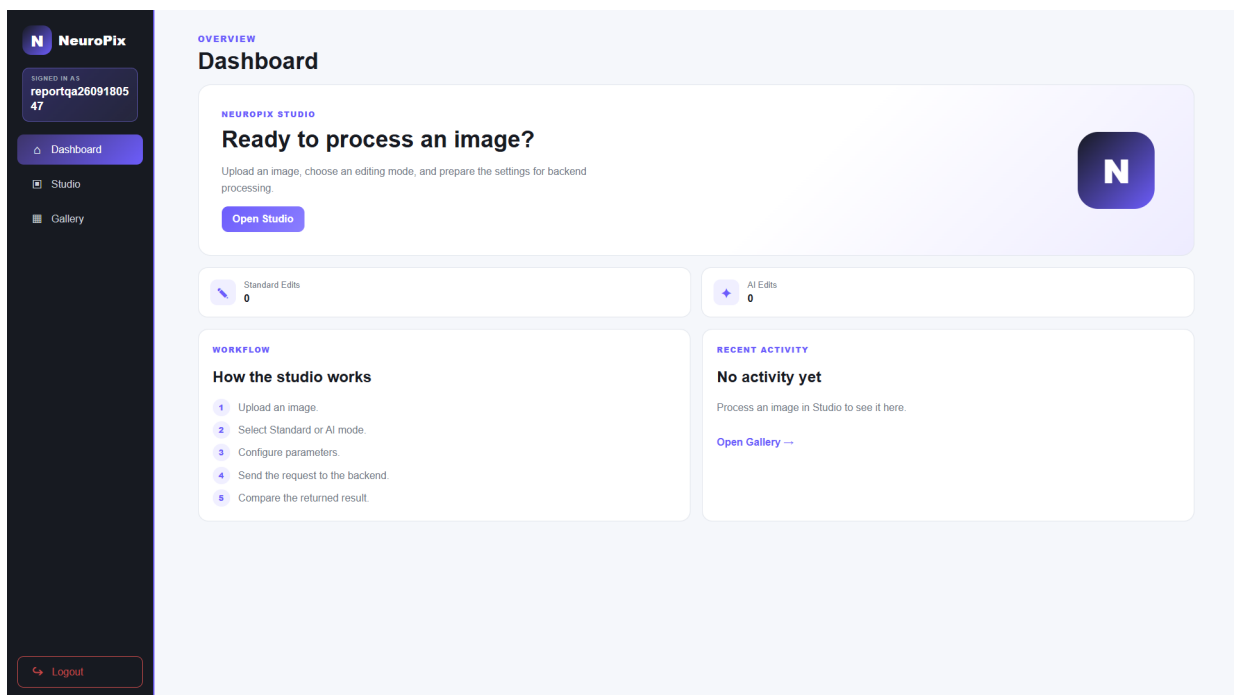
Figure 6.3: UC-T02 login and protected-access sequence.

Table 6.4: UC-T02 execution result.

Step	Observed result
Enter credentials	Username displayed and password remained masked
Submit login	Dashboard opened and showed the signed-in username, private navigation, and empty initial counts
Final status	Pass



UC-T02 (a) Login credentials entered



UC-T02 (b) Authenticated dashboard

Figure 6.4: UC-T02 login evidence.

6.6 UC-T03: Upload and Apply a Standard Edit

Precondition: UC-T02 has established the session. The user opens the Studio and uploads the controlled image. Cropping, rotation, and maximum grayscale make the changes easy to see. The expected result is a processed image available for comparison.

UC-T03 Standard-edit sequence

Precondition: signed in; this test uploads the source used by the following AI tests

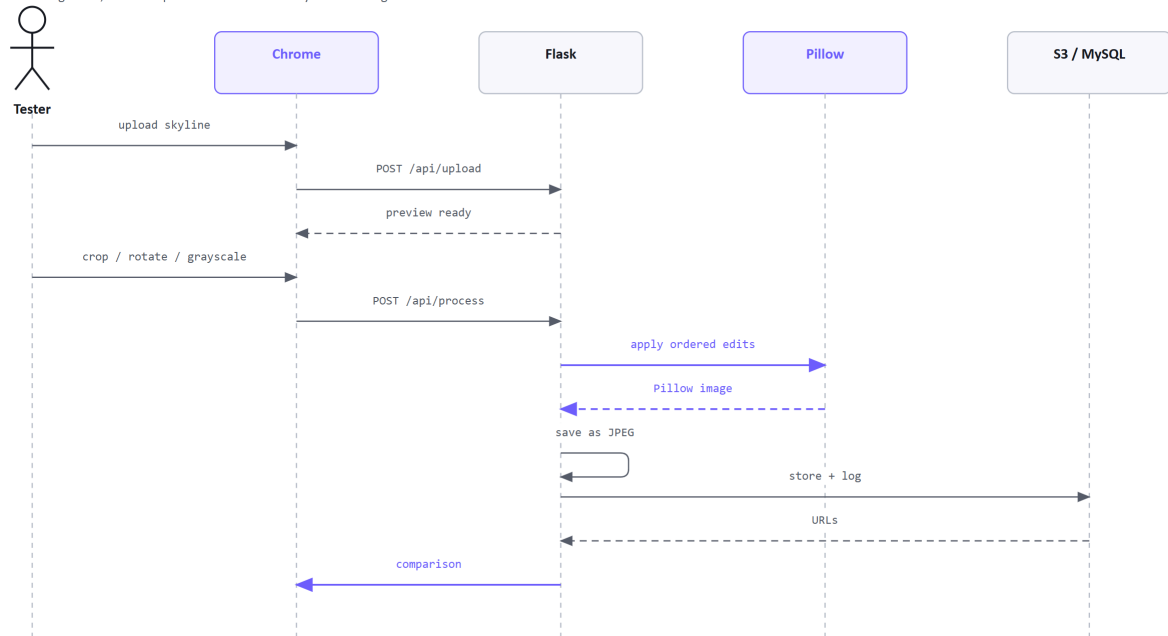
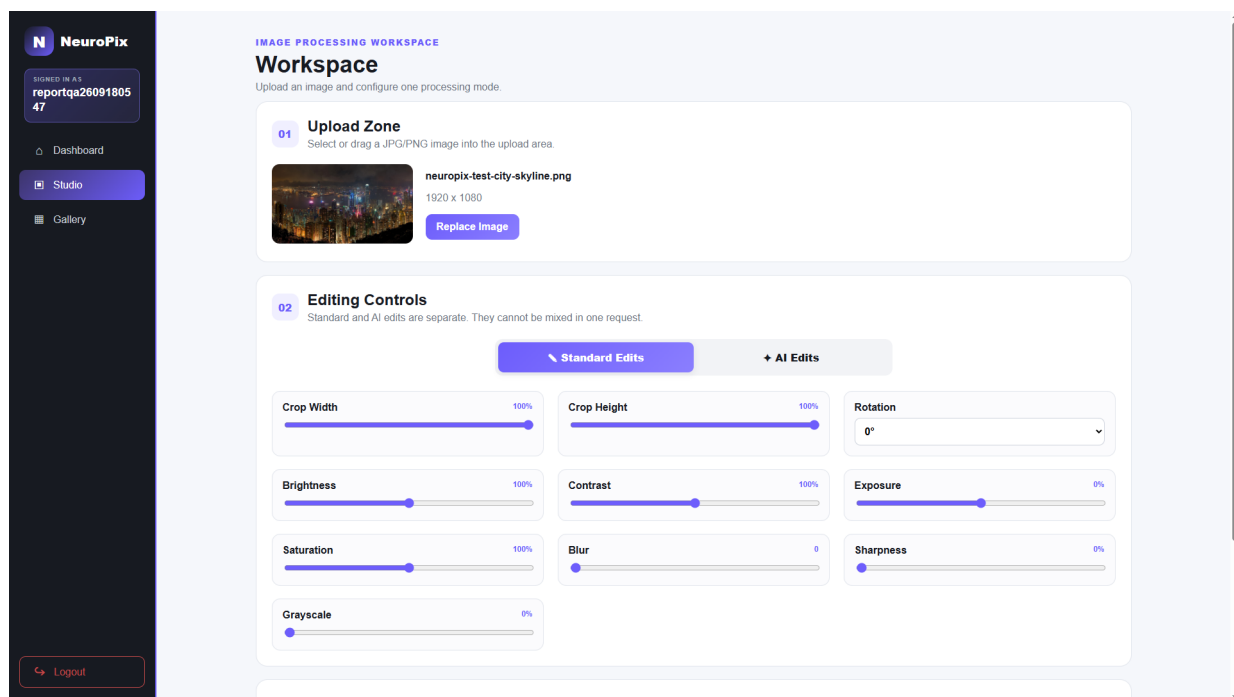


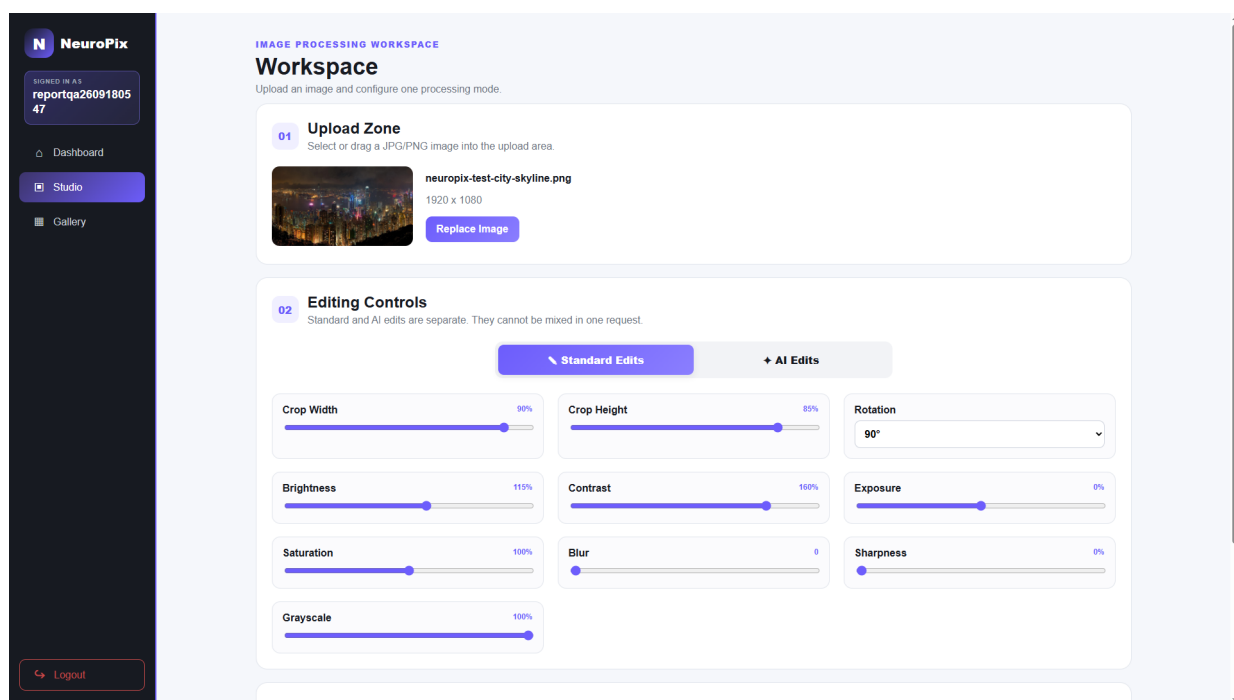
Figure 6.5: UC-T03 upload and Standard-edit sequence.

Table 6.5: UC-T03 execution result.

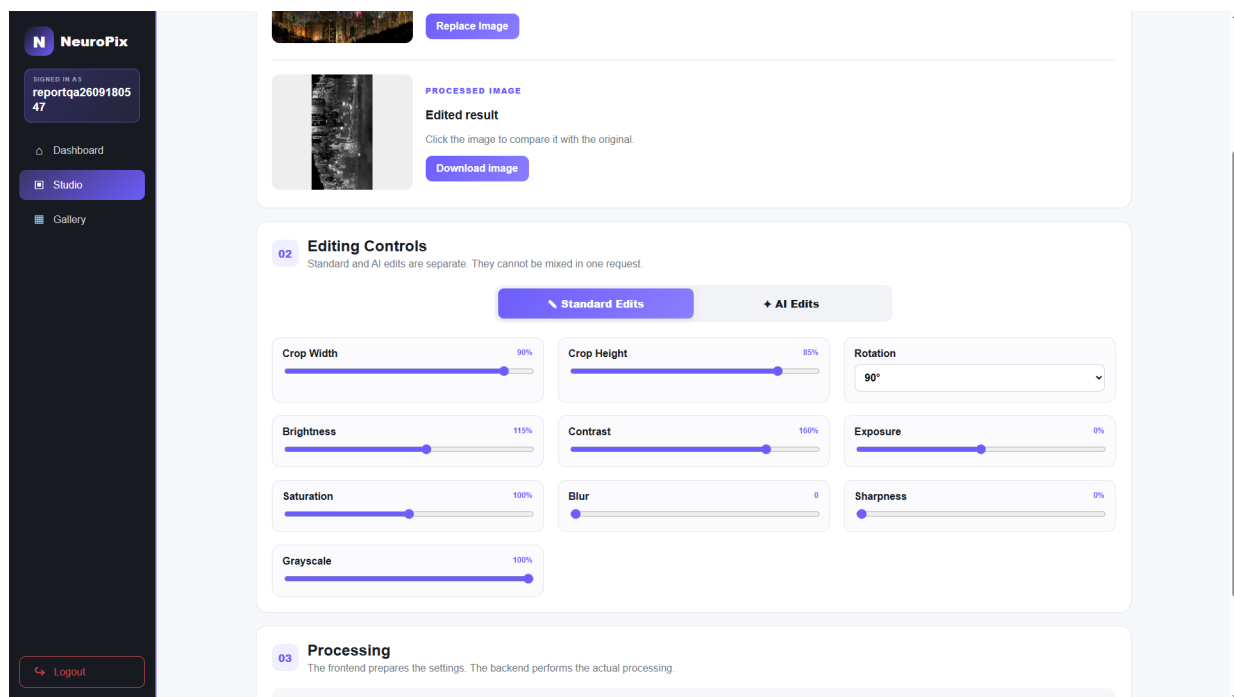
Step	Observed result
Upload	The source preview displayed the correct filename and 1920 × 1080 dimensions
Configure	All selected values appeared beside their controls, including 100% grayscale
Process	A processed preview and success message appeared
Compare	The modal showed the original beside the rotated, cropped, high-contrast grayscale result
Final status	Pass



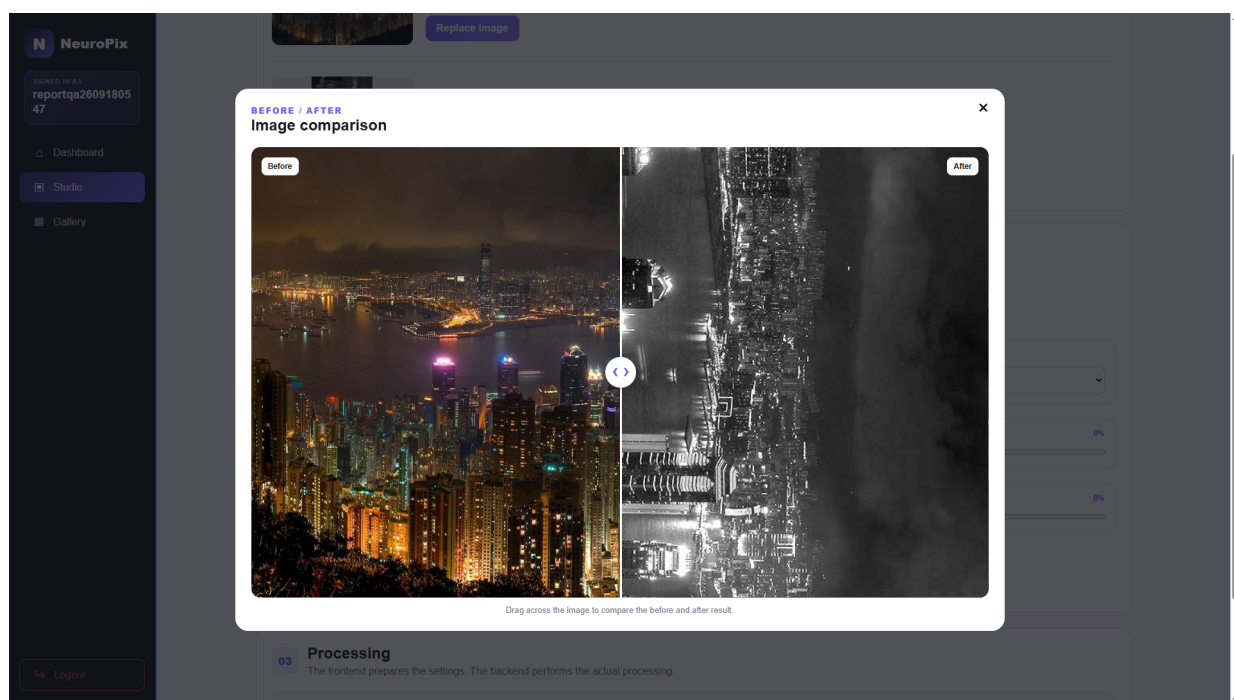
UC-T03 (a) City skyline uploaded



UC-T03 (b) Standard settings selected



UC-T03 (c) Processed result



UC-T03 (d) Before-and-after comparison

Figure 6.6: UC-T03 Standard-edit evidence.

6.7 UC-T04: Edit with OpenAI

Precondition: the user is authenticated and the controlled image is staged. The user selects OpenAI and enters content, background, and enhancement instructions. The expected result is a new image available for comparison, with no repeated login or upload.

UC-T04 OpenAI-edit sequence

Precondition: signed in with the source already staged; no repeated upload

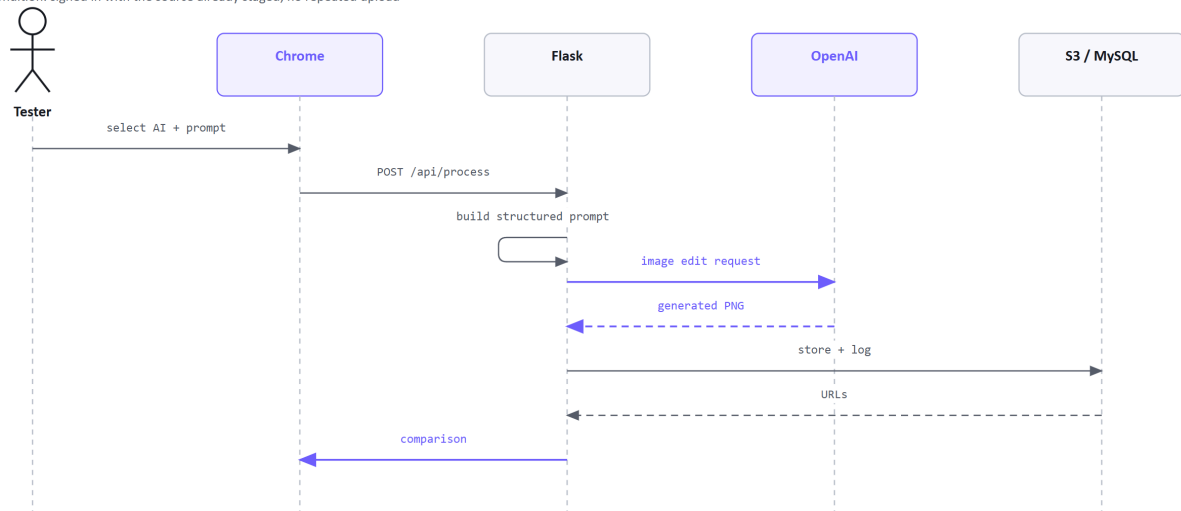
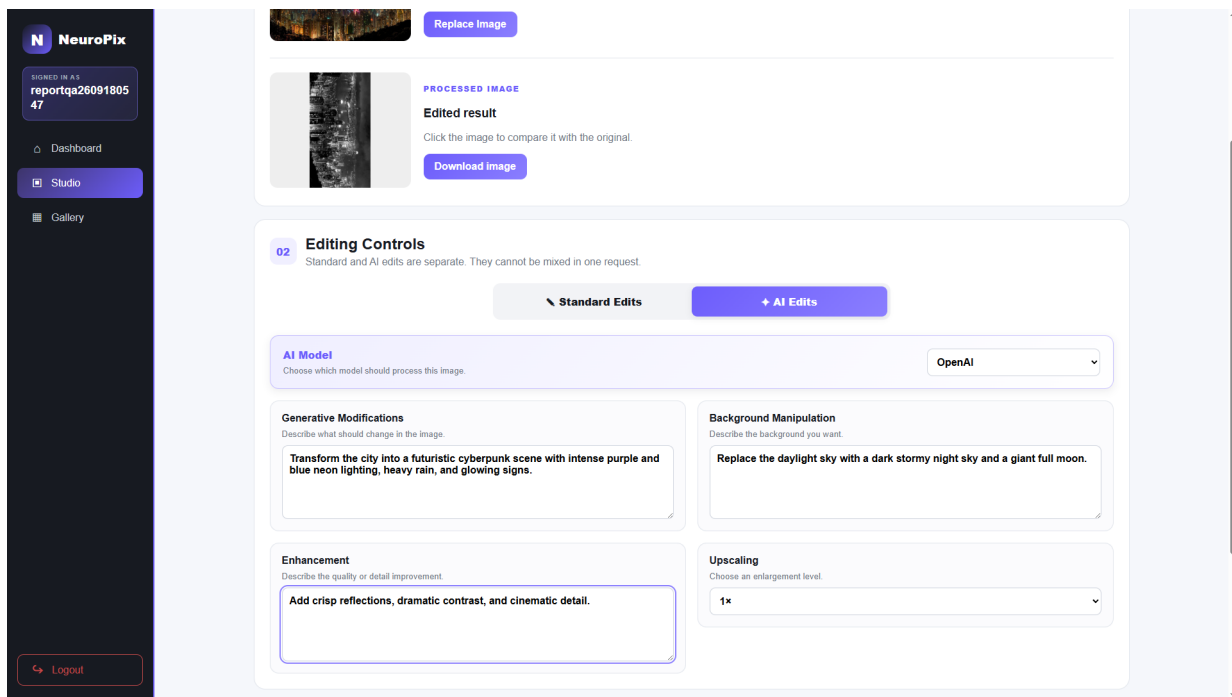


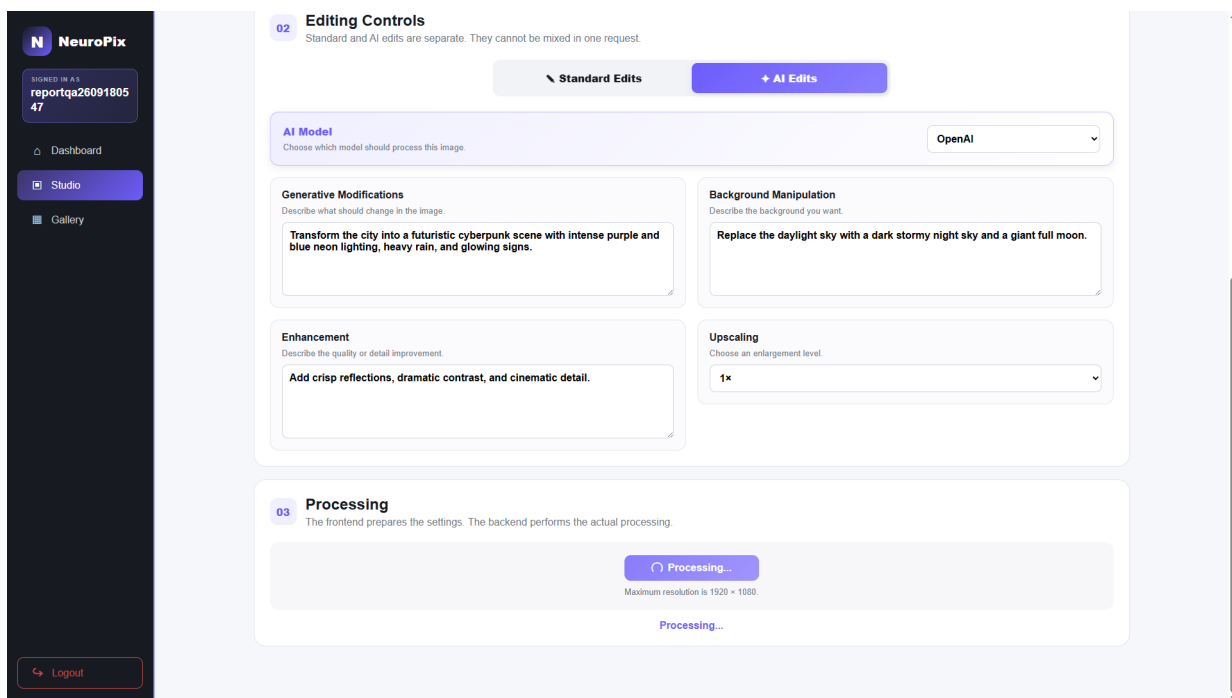
Figure 6.7: UC-T04 OpenAI-edit sequence.

Table 6.6: UC-T04 execution result.

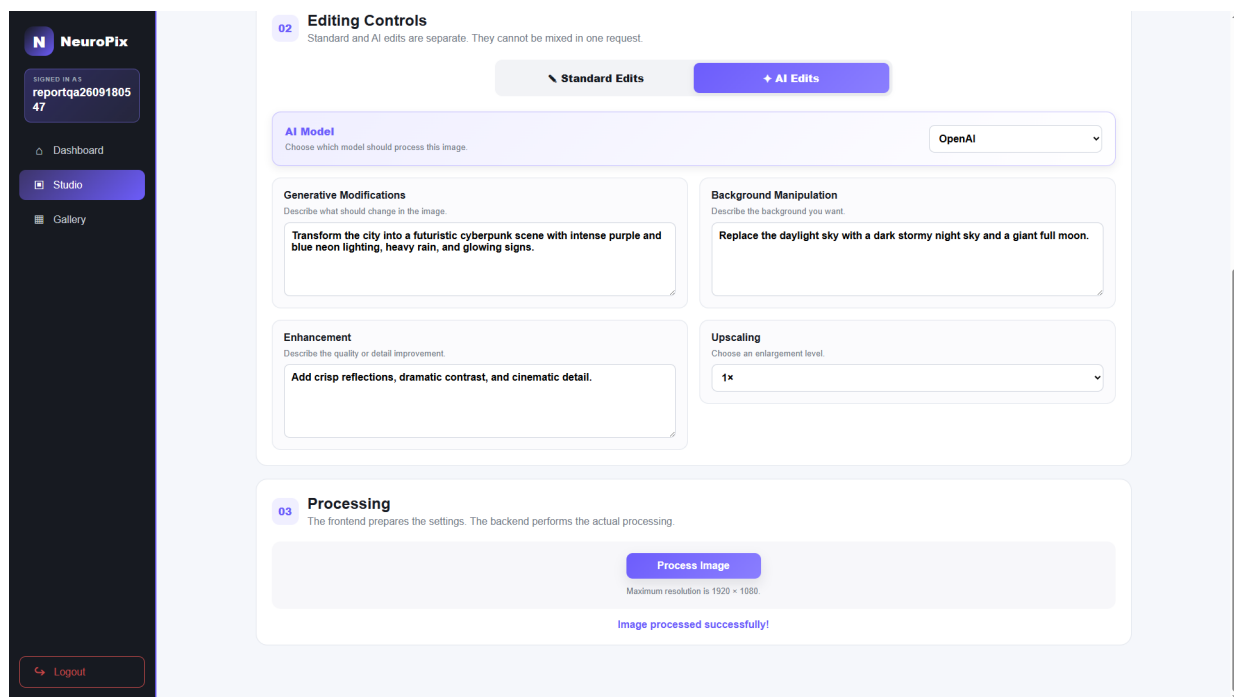
Step	Observed result
Configure	OpenAI was the selected provider and all three purpose-specific fields retained the instructions
Submit	The Process button changed to a disabled processing state
Return result	The success state appeared and the processed preview was replaced
Compare	The result showed an obvious neon night city with rain, a large moon, and altered lighting while retaining the broad skyline composition
Final status	Pass



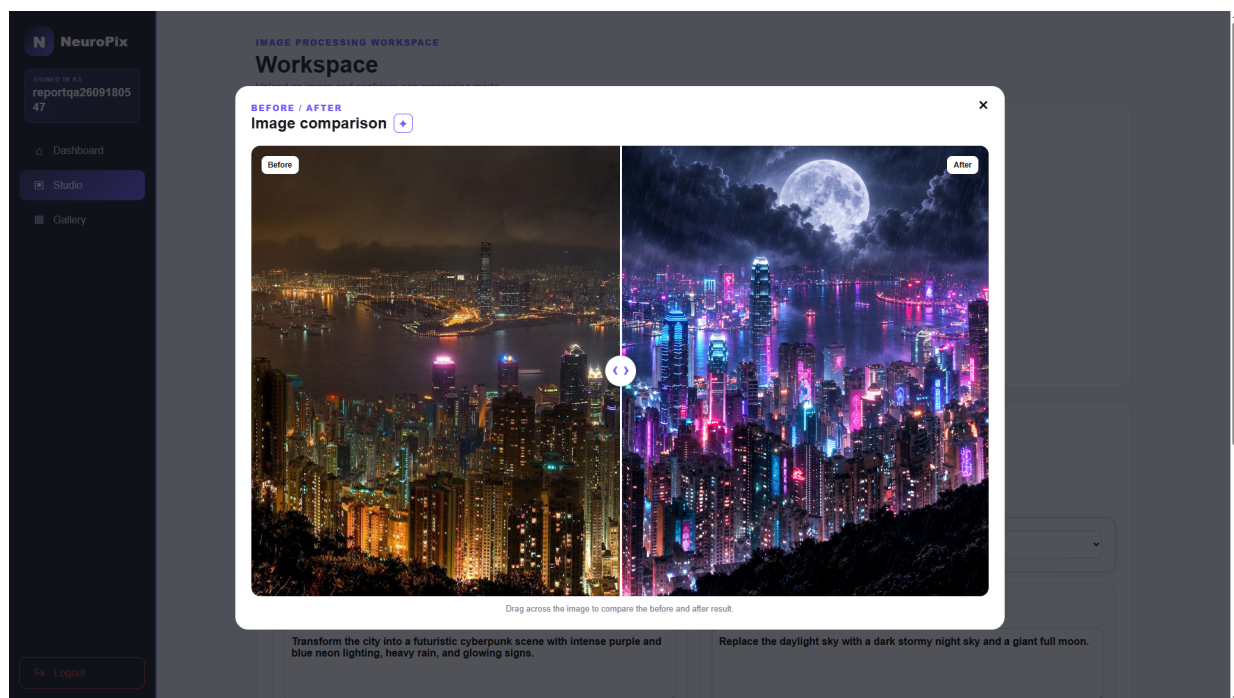
UC-T04 (a) OpenAI instructions entered



UC-T04 (b) OpenAI processing



UC-T04 (c) OpenAI result returned



UC-T04 (d) Before-and-after comparison

Figure 6.8: UC-T04 OpenAI-edit evidence.

6.8 UC-T05: Edit with Local Stable Diffusion

Precondition: the user is authenticated, the controlled source is staged, and the local computer, Docker service, and reverse SSH tunnel are active. The availability check must report the local provider as ready. The user selects Local Stable Diffusion and submits the same instructions used for OpenAI. The expected result is a new image available for comparison.

UC-T05 local Stable Diffusion sequence

Precondition: signed in, source staged, and local service plus tunnel ready

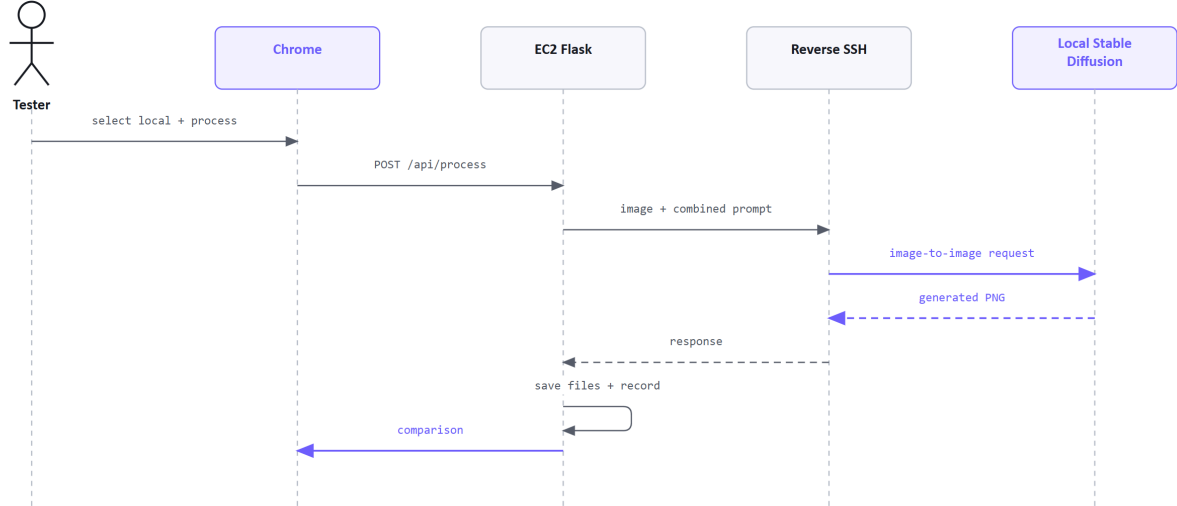
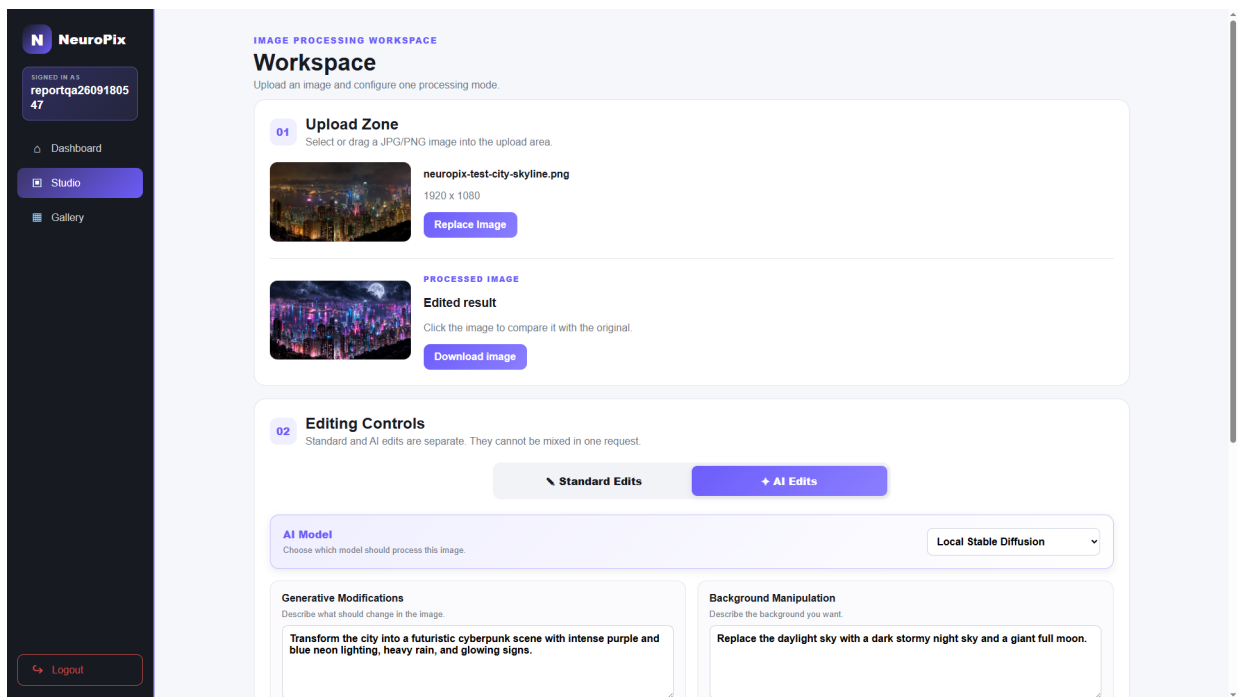


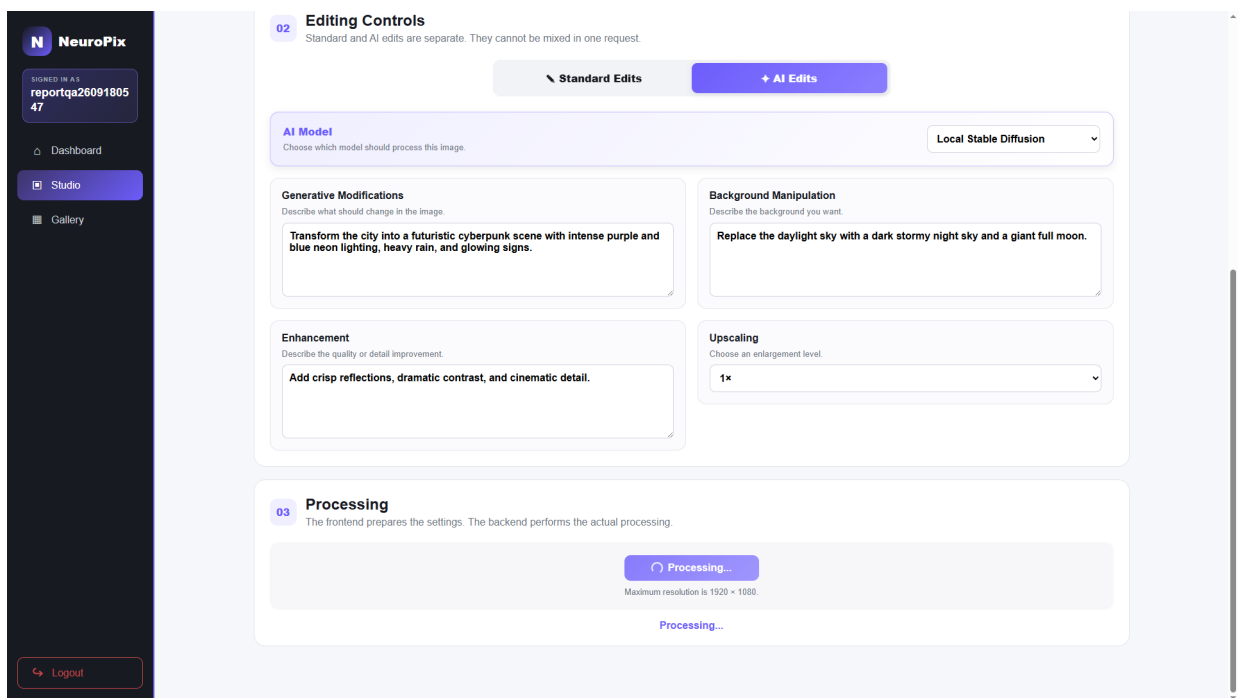
Figure 6.9: UC-T05 local Stable Diffusion sequence.

Table 6.7: UC-T05 execution result.

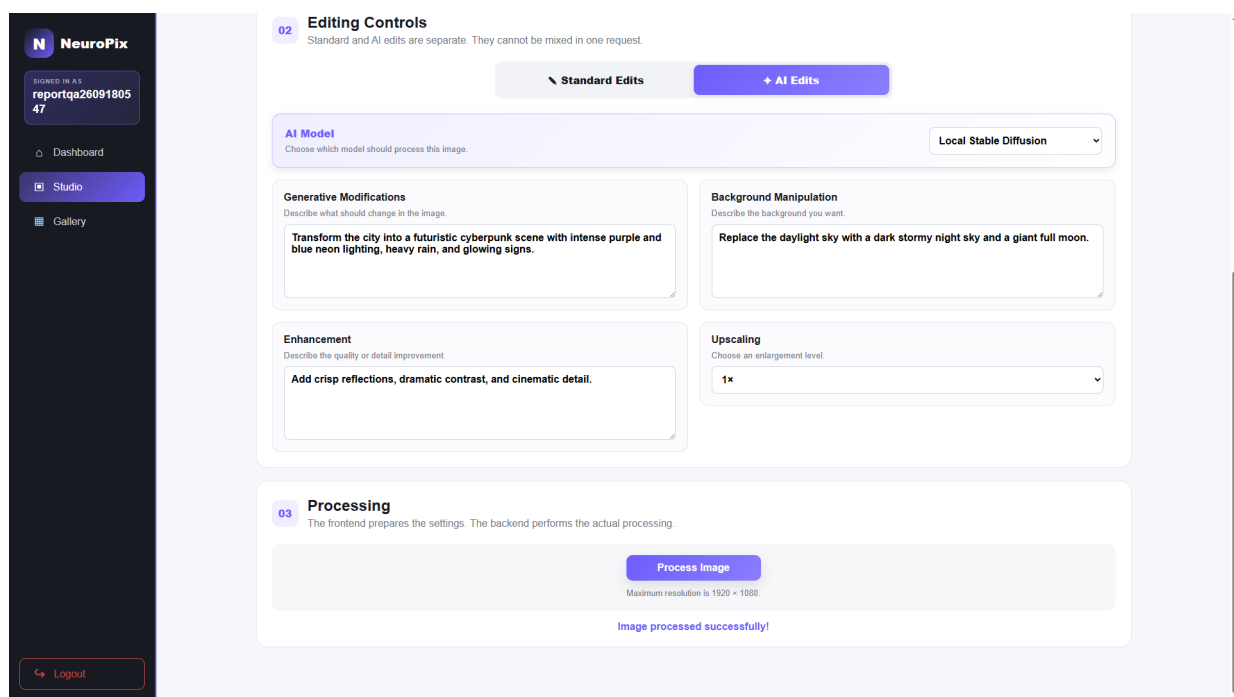
Step	Observed result
Select provider	Local Stable Diffusion appeared as the active provider
Submit	The request entered the disabled processing state while the local service generated the image
Return result	The site reported success and displayed a new processed preview
Compare	The local result produced a strong purple futuristic city transformation that was visibly different from both the source and hosted result
Final status	Pass



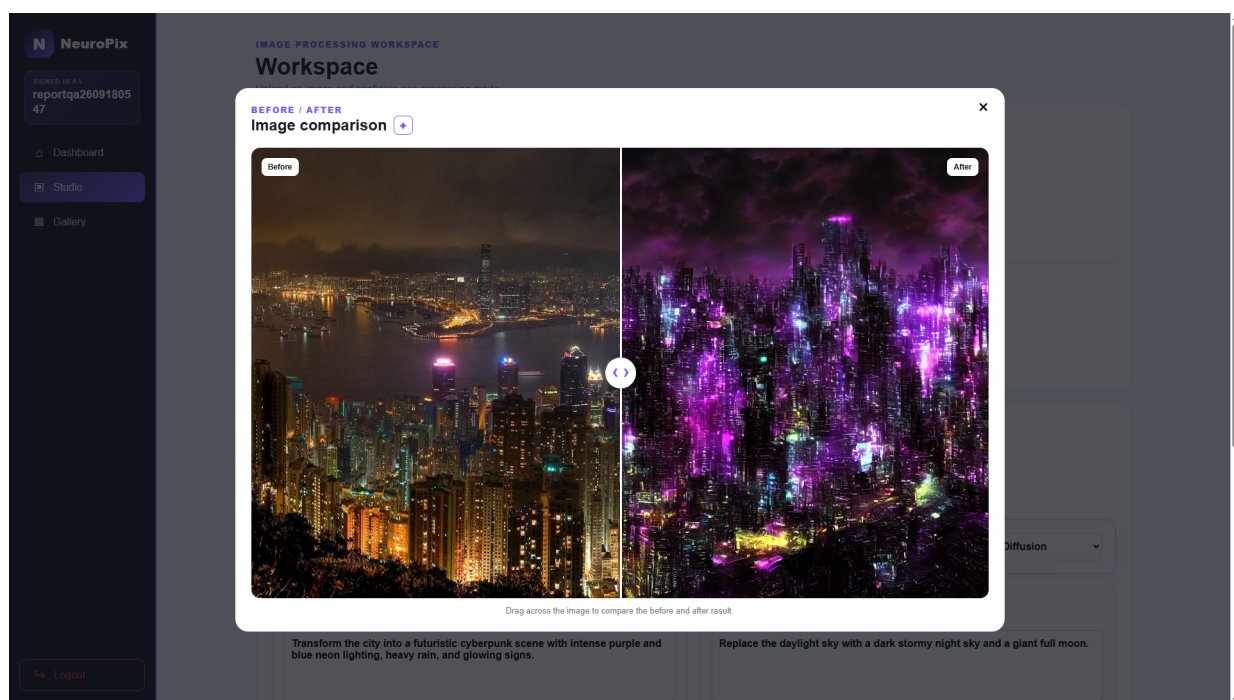
UC-T05 (a) Local provider selected



UC-T05 (b) Local processing



UC-T05 (c) Local result returned



UC-T05 (d) Before-and-after comparison

Figure 6.10: UC-T05 local-model evidence.

6.9 UC-T06: Verify and Delete a Gallery Image

Precondition: UC-T03, UC-T04, and UC-T05 have created saved results for the current user. The user verifies the latest gallery image, confirms deletion, and refreshes the page. The selected image should remain absent while the other saved results remain visible.

UC-T06 gallery verification and deletion sequence

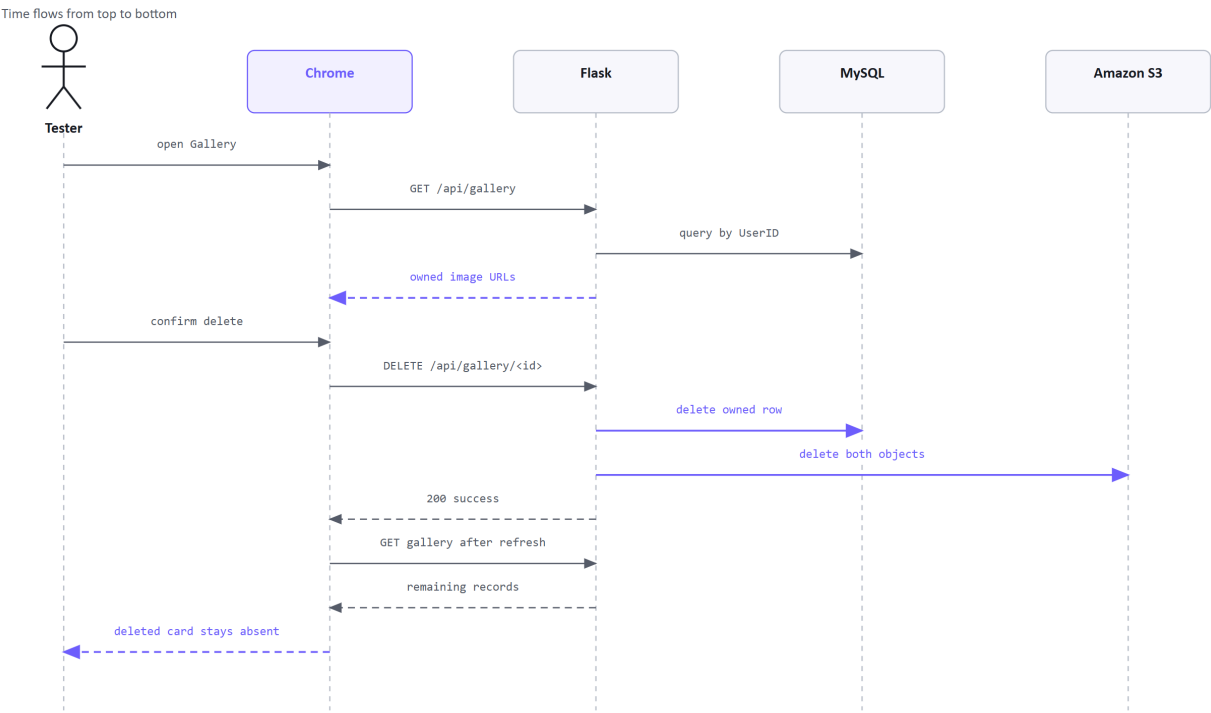
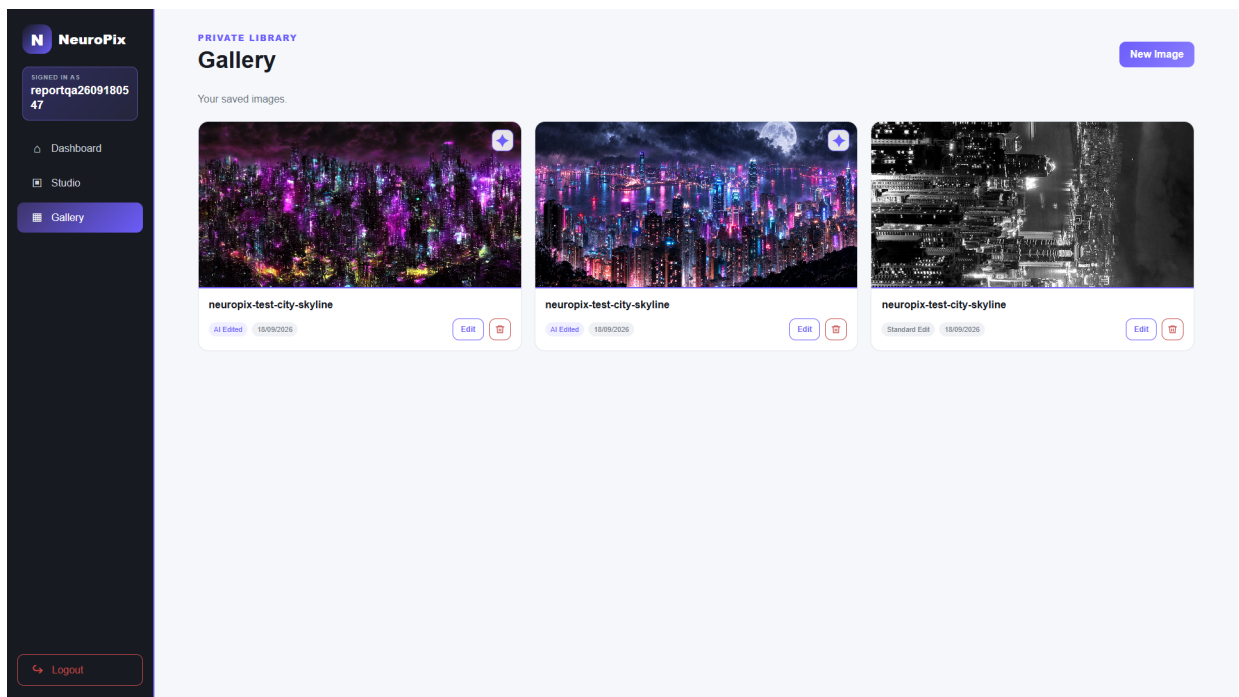


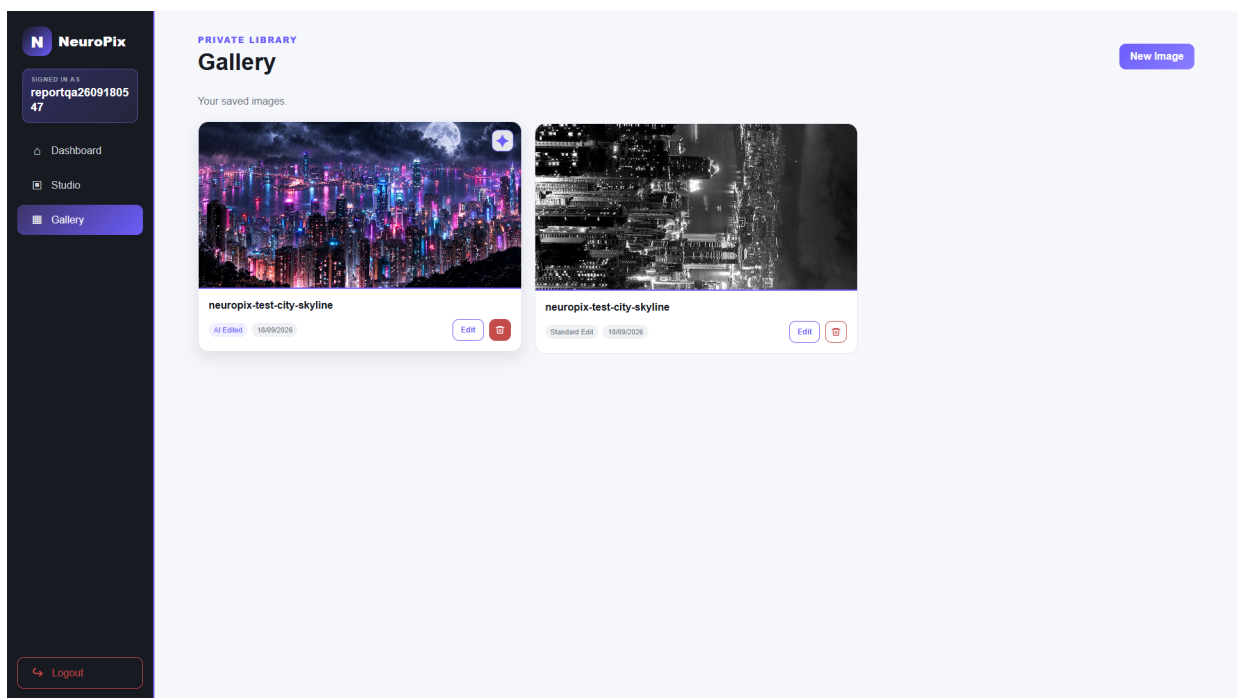
Figure 6.11: UC-T06 gallery verification and deletion sequence.

Table 6.8: UC-T06 execution result.

Step	Observed result
Open gallery	Three saved results appeared: two AI edits and one Standard edit
Delete latest result	Confirmation was accepted and the selected card was removed
Verify refresh	Two expected cards remained and no deleted card reappeared
Final status	Pass



UC-T06 (a) Saved image present in gallery



UC-T06 (b) Gallery after deletion

Figure 6.12: UC-T06 gallery evidence.

6.10 Results Summary and Evaluation

Table 6.9: Consolidated verification results.

Test	Scope	Result
Automated API and security checks	28 API, password, local-provider, and mocked-persistence tests	Pass
UC-T01	Account registration	Pass
UC-T02	Login and protected dashboard	Pass
UC-T03	Upload, Standard processing, and comparison	Pass
UC-T04	Structured OpenAI edit and comparison	Pass
UC-T05	Local Stable Diffusion edit through the active tunnel	Pass
UC-T06	Gallery presence and deletion	Pass

All six live cases passed in the stated production environment. Standard editing produced the selected crop, rotation, and grayscale changes. Both AI providers returned transformed images, and gallery deletion remained visible after refresh. These results support the integrated workflow, not a ranking of model quality. The AI cases used one source and one set of instructions. Download and gallery re-editing were not covered by separate live cases. Performance under load, long-term reliability, comparative usability, and repeatability of AI output remain unmeasured.

Chapter 7 Conclusion and Future Work

7.1 Achievement of the Project Aim

NeuroPix met its development aim by delivering a deployed web application for Standard and AI-assisted editing, comparison, download, and personal image management. The live tests supported the core editing and gallery workflows, while the automated checks covered selected backend behavior. We integrated these features within our budget and available hardware. Production-scale reliability and comparative usability still need to be measured.

7.2 Critical Evaluation and Lessons Learned

A shared processing route allowed the same interface to support hosted and local AI. Their operating needs remained different, so the deployment had to account for both an external service and a computer maintained by the team.

The main trade-off was operational simplicity against resilience. Synchronous requests were straightforward to integrate, but leave users waiting for AI results. Hosting the local model on equipment we owned reduced cloud GPU costs, while making availability depend on that computer and its tunnel. Coordinating S3 and MySQL also introduced partial-failure risks that the current implementation does not fully resolve.

Working directly on main made Git easier for inexperienced teammates. Student offers and the RTX 3080 we already owned helped control costs. A larger team would benefit from branches, review, and a more managed deployment to isolate changes and reduce operational dependencies.

We learned to check the application's behavior against our plans and to look beyond one provider when student offers did not cover our needs. Visual comparison was also necessary to judge whether an AI edit was useful, even when the request completed successfully.

7.3 Recommended Six-Month Continuation

For another six months, we would tackle recovery and safer deployment first, then improve processing and evaluate the results. The priorities are:

- Improve recovery when image storage or database updates fail.
- Adopt feature branches, review, and staging.
- Move AI work to a recoverable background queue with progress reporting.
- Improve local output quality within RTX 3080 limits.
- Measure performance, accessibility, and cross-browser behavior.

7.4 Final Conclusion

NeuroPix brings Standard editing, guided AI requests, comparison, and saved results into one browser workspace. The project demonstrates that this workflow can be delivered with our available hardware and budget. The next priority is dependable operation and a measured evaluation of its usefulness to users.

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